

SOLVE-IT Workshop

DFRWS USA 2025, Chicago

Dr Chris Hargreaves

Lecturer, Department of Computer Science, University of Oxford

Director, HARGS Solutions Ltd.

Co-Founder, SOLVE-IT Knowledge Base



SOLVE-IT Topics

- SOLVE-IT Structure
- SOLVE-IT Report generation
- SOLVE-IT Potential tool integration
- SOLVE-IT Contributing

SOLVE-IT structure

Background

- The project is heavily influenced by MITRE ATT&CK

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection
10 techniques	8 techniques	10 techniques	14 techniques	20 techniques	14 techniques	44 techniques	17 techniques	32 techniques	9 techniques	17 techniques
Active Scanning (3)	Acquire Access	Content Injection	Cloud Administration Command	Account Manipulation (7)	Abuse Elevation Control Mechanism (6)	Abuse Elevation Control Mechanism (6)	Adversary-in-the-Middle (4)	Account Discovery (4)	Exploitation of Remote Services	Adversary-in-the-Middle (4)
Gather Victim Host Information (4)	Acquire Infrastructure (8)	Drive-by Compromise	Command and Scripting Interpreter (11)	BITS Jobs	Access Token Manipulation (5)	Access Token Manipulation (5)	Brute Force (4)	Application Window Discovery	Internal Spearphishing	Archive Collection Data (4)
Gather Victim Identity Information (3)	Compromise Accounts (3)	Exploit Public-Facing Application	Container Administration Command	Boot or Logon Autostart Execution (14)	Account Manipulation (7)	BITS Jobs	Credentials from Password Stores (6)	Browser Information Discovery	Lateral Tool Transfer	Audio Collection (4)
Gather Victim Network Information (6)	Compromise Infrastructure (8)	External Remote Services	Deploy Container	Boot or Logon Initialization Scripts (5)	Boot or Logon Autostart Execution (14)	Build Image on Host	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking (2)	Automated Collection (4)
Gather Victim Org Information (4)	Develop Capabilities (4)	Hardware Additions	Exploitation for Client Execution	Browser Extensions	Boot or Logon Initialization Scripts (5)	Debugger Evasion	Forced Authentication	Cloud Service Dashboard	Remote Services (8)	Browsing Session Hijacking (4)
Phishing for Information (4)	Establish Accounts (3)	Phishing (4)	Inter-Process Communication (3)	Compromise Host Software Binary	Create or Modify System Process (5)	Deobfuscate/Decode Files or Information	Forge Web Credentials (2)	Cloud Service Discovery	Replication Through Removable Media	Clipboard Collection (4)
Search Closed Sources (2)	Obtain Capabilities (7)	Replication Through Removable Media	Native API	Create Account (3)	Domain or Tenant Policy Modification (2)	Deploy Container	Input Capture (4)	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud (4)
Search Open Technical Databases (5)	Stage Capabilities (6)	Supply Chain Compromise (3)	Scheduled Task/Job (5)	Create or Modify System Process (5)	Escape to Host	Direct Volume Access	Modify Authentication Process (9)	Container and Resource Discovery	Taint Shared Content	Data from Configuration Repositories (4)
Search Open Websites/Domains (3)		Trusted Relationship	Serverless Execution	Event Triggered Execution (17)	Event Triggered Execution (17)	Domain or Tenant Policy Modification (2)	Multi-Factor Authentication Interception	Debugger Evasion		Data from Information Repositories (4)
Search Victim-Owned Websites			Shared Modules			Execution Guardrails (2)		Device Driver Discovery		Data from Information Repositories (4)
						Exploitation for Defense Evasion		Domain Trust Discovery		Data from Information Repositories (4)
						File and Directory Permissions				Data from Information Repositories (4)

Drive-by Compromise

Adversaries may gain access to a system through a user visiting a website over the normal course of browsing. With this technique, the user's web browser is typically targeted for exploitation, but adversaries may also use compromised websites for non-exploitation behavior such as acquiring [Application Access Token](#).

Multiple ways of delivering exploit code to a browser exist (i.e., [Drive-by Target](#)), including:

- A legitimate website is compromised where adversaries have injected some form of malicious code such as JavaScript, iFrames, and cross-site scripting
- Script files served to a legitimate website from a publicly writeable cloud storage bucket are modified by an adversary
- Malicious ads are paid for and served through legitimate ad providers (i.e., [Malvertising](#))
- Built-in web application interfaces are leveraged for the insertion of any other kind of object that can be used to display web content or contain a script that executes on the visiting client (e.g. forum posts, comments, and other user controllable web content).

Often the website used by an adversary is one visited by a specific community, such as government, a particular industry, or region, where the goal is to compromise a specific user or set of users based on a shared interest. This kind of targeted campaign is often referred to a strategic

ID: T1189

Sub-techniques: No sub-techniques

ⓘ

Tactic: [Initial Access](#)

ⓘ

Platforms: Identity Provider, Linux, Windows, macOS

Contributors: Jeff Sakowicz, Microsoft Identity Developer Platform Services (IDPM Services); Saisha Agrawal, Microsoft Threat Intelligent Center (MSTIC)

Version: 1.6

Created: 18 April 2018

Last Modified: 15 October 2024

[Version](#) [Permalink](#)

Initial Access	Discovery	Lateral Movement	Collection
Techniques	32 techniques	9 techniques	17 techniques
Account Discovery (4)	Account Discovery (4)	Exploitation of Remote Services	Adversary Impersonation
Application Window Discovery	Application Window Discovery	Internal Spearphishing	Archiving Collected Data (2)
Browser Information Discovery	Browser Information Discovery	Lateral Tool Transfer	Audio Collection
Cloud Infrastructure Discovery	Cloud Infrastructure Discovery	Remote Service Session Hijacking (2)	Automated Collection
Cloud Service Dashboard	Cloud Service Dashboard	Remote Services (8)	Browsing Session Hijacking
Cloud Service Discovery	Cloud Service Discovery	Replication Through Removable Media	Clipboard Collection
Cloud Storage Object Discovery	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud Repositories
Container and Resource Discovery	Container and Resource Discovery	Taint Shared Content	Data from Configuration Repositories
Debugger Evasion	Debugger Evasion		Data from Information Repositories
Device Driver Discovery	Device Driver Discovery		Data from Information Repositories
Domain Trust Discovery	Domain Trust Discovery		Data from Information Repositories

Also has examples, mitigations, detection etc.

Background

Mitigations

ID	Mitigation	Description
M1048	Application Isolation and Sandboxing	Browser sandboxes can be used to mitigate some of the impact of exploitation, but sandbox escapes may still exist. ^{[68][69]} Other types of virtualization and application microsegmentation may also mitigate the impact of client-side exploitation. The risks of additional exploits and weaknesses in implementation may still exist for these types of systems. ^[69]
M1050	Exploit Protection	Security applications that look for behavior used during exploitation such as Windows Defender Exploit Guard (WDEG) and the Enhanced Mitigation Experience Toolkit (EMET) can be used to mitigate some exploitation behavior. ^[70] Control flow integrity checking is another way to potentially identify and stop a software exploit from occurring. ^[71] Many of these protections depend on the architecture and target application binary for compatibility.
M1021	Restrict Web-Based Content	For malicious code served up through ads, adblockers can help prevent that code from executing in the first place. Script blocking extensions can help prevent the execution of JavaScript that may commonly be used during the exploitation process.
M1051	Update Software	Ensure all browsers and plugins kept updated can help prevent the exploit phase of this technique. Use modern browsers with security features turned on.

Detection

ID	Data Source	Data Component	Detects
M1001	...	...	...

Initial Access	Discovery	Lateral Movement	Collection
17 techniques	32 techniques	9 techniques	17 techniques
...	Account Discovery (4)	Exploitation of Remote Services	Adversary ...
...	Application Window Discovery	Internal Spearphishing	Archiv ...
...	Browser Information Discovery	Lateral Tool Transfer	Audio ...
...	Cloud Infrastructure Discovery	Remote Service Session Hijacking (2)	Autom ...
...	Cloud Service Dashboard	Remote Services (8)	Brows ...
...	Cloud Service Discovery	Replication Through Removable Media	Session Hijack
...	Cloud Storage Object Discovery	Software Deployment Tools	Clipboard
...	Container and Resource Discovery	Taint Shared Content	Data f ...
...	Debugger Evasion	Domain Trust Discovery	Data f ...
...	Device Driver Discovery		Data f ...
...	Domain Trust Discovery		Data f ...

Can we construct something similar for digital forensics and is it useful?

SOLVE-IT structure: overview

[illegible]

SOLVE-IT structure: overview

[illegible]

SOLVE-IT structure: overview

[illegible]

SOLVE-IT structure: overview

[illegible]

SOLVE-IT structure: overview

Find potential evidence sources	Prioritise	Preserve	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Extract Partition and File System Information	Extract Operating System Feature Information	Extract Application-based Information	Examine data at the file-level	Locate Relevant Digital Artefacts	Establish Identities	Create Visualisation	Reconstruct Events	Research	Documentation
Crime scene searching T1005	Triage T1001	Place device in faraday environment	Disk imaging T1002	Key recovery from memory	Disk image hash verification	Privileged material protection	Identify partitions T1059	Content indexer examination (OS)	Browser examination T1069	Database examination T1071	Keyword searching T1049	Extraction of user accounts T1084	Virtualise suspect system for previewing T1103	Location-based event reconstruction T1087	Source code review T1089	Bookmarking T1091
Technique name:	Disk imaging					back to main										
Technique ID:	T1002															
Category:	['Acquire']															
Description:	Copying of sectors from a storage media, typically LBA0 to LBAmx into an imaging format. The could be from a traditional hard disk, SSD, USB stick, or data from an eMMC chip that has been desoldered and placed in a reader.															
Synonyms:	[]															
Details:																
Subtechniques:	[]															
CASE output entities:	['observable:Image']															
Examples:	['dcfldd', 'FTK Imager', 'Magnet ACQUIRE']															
Potential weaknesses:																
Weakness ID:	Detail:					INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISINT	Mitigations				
W1004	Acquisition does not include all sectors from LBA0 to LBA max					x						M1003,M1004,				
W1006	Acquisition does not include data in HPA					x						M1005,				
W1007	Acquisition does not include data in DCO					x						M1006,				
W1013	Acquisition includes extra bytes						x					M1003,M1009,				
W1014	Imaging process changes original data								x			M1007,M1008,				
W1015	Powering on SSD results in sectors being wiped by TRIM operation					x			x	x						
W1016	Data copied from sectors on source are stored incorrectly								x	x		M1009,M1004,				
W1136	Not recovering data from a failed hard drive					x						M1089,				
W1143	Acquisition method does not read remapped sectors e.g. G-Lists					x						M1102,				
Mitigations:																
M1003	Check image size corresponds with drive label															
M1004	Check hash of image matches hash of source device															
M1005	Testing to ensure software and hardware setup detects HPAs															
M1006	Testing to ensure software and hardware setup detects DCOs															
M1007	Use hardware write blocker (T1012)															
M1008	Use software write blocker (T1013)															
M1009	Check hash of image matches hash of source material															
M1089	Attempt physical disk repair															
M1102	Apply techniques to read remapped sectors															
References:	Nikkel, B., 2016. Practical forensic imaging: securing digital evidence with Linux tools. No Starch Press, Chapter 6, “Forensic Image											['T1002']				
	Gupta, M.R., Hoeschele, M.D. and Rogers, M.K., 2006. Hidden disk areas: HPA and DCO. International Journal of Digital Evidence, 5(1),											['W1006', 'W1007']				
	DSTL Digital Forensics Bulletin, TRIM & Garbage Collection on SSDs, Edition 27, July 2024 DSTL/PUB160897,											['W1015']				
	Turner, P. The Remapped/Reallocated Sector Conundrum, Presentation at DFRWS US 2024, https://www.dfrws.org/wp-											['W1143', 'M1102']				

	Synonyms:	[]									Documentation
	Details:										Bookmarking T1091
Crime scene search T1005	Subtechniques:	[]									Increase bookmark-based automated report T1092
Digital sniffer detection T1006	CASE output entities:	['observable:Image']									Write expert report T1093
	Examples:	['dcfldd', 'FTK Imager', 'Magnet ACQUIRE']									
	Potential weaknesses:										
	Weakness ID:	Detail:	INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISINT	Mitigations		
Profiling network T1008	W1004	Acquisition does not include all sectors from LBA0 to LBA max	x						M1003,M1004,		Disclosure T1094
	W1006	Acquistion does not include data in HPA	x						M1005,		
Locate cloud acc identifiers T1009	W1007	Acquistion does not include data in DCO	x						M1006,		
	W1013	Acquisition includes extra bytes		x					M1003,M1009,		
	W1014	Imaging process changes original data				x			M1007,M1008,		
	W1015	Powering on SSD results in sectors being wiped by TRIM operation	x			x	x				
	W1016	Data copied from sectors on source are stored incorrectly				x	x		M1009,M1004,		
	W1136	Not recovering data from a failed hard drive	x						M1089,		
	W1143	Acquisition method does not read remapped sectors e.g. G-Lists	x						M1102,		
	Mitigations:										
	M1003	Check image size corresponds with drive label									
	M1004	Check hash of image matches hash of source device									
	M1005	Testing to ensure software and hardware setup detects HPAs									
	M1006	Testing to ensure software and hardware setup detects DCOs									
	M1007	Use hardware write blocker (T1012)									
	M1008	Use software write blocker (T1013)									
	M1009	Check hash of image matches hash of source material									
	M1089	Attempt physical disk repair									
	M1102	Apply techniques to read remapped sectors									
	References:	Nikkel, B., 2016. Practical forensic imaging: securing digital evidence with Linux tools. No Starch Press, Chapter 6, “Forensic Image							['T1002']		
		Gupta, M.R., Hoeschele, M.D. and Rogers, M.K., 2006. Hidden disk areas: HPA and DCO. International Journal of Digital Evidence, 5(1),							['W1006', 'W1007']		
		DSTL Digital Forensics Bulletin, TRIM & Garbage Collection on SSDs, Edition 27, July 2024 DSTL/PUB160897,							['W1015']		
		Turner, P. The Remapped/Reallocated Sector Conundrum, Presentation at DFRWS US 2024, https://www.dfrws.org/wp-							['W1143', 'M1102']		

SOLVE-IT structure: overview

- On release:
 - 104 techniques
 - 17 categories
 - 33 populated
 - 3 community contributors
 - 156 weaknesses identified
 - 108 mitigations indexed

[illegible]

SOLVE-IT structure: overview

- Now:
 - ~~104~~ 117 techniques
 - 17 categories
 - ~~33~~ 45 populated
 - ~~3~~ >10 community contributors
 - ~~156~~ 188 weaknesses identified
 - ~~108~~ 137 mitigations indexed

[illegible]

SOLVE-IT structure: implementation

- Hosted in GitHub:

The screenshot shows the GitHub repository for SOLVE-IT. The repository is public and has 39 stars, 5 forks, and 7 watchers. It is managed by chrishargreaves. The repository contains a README, a MIT license, and a list of files including .github, .repo_info, data, lab_config_examples, reporting_scripts, solve_it_library, tests, .gitignore, CONTRIBUTING.md, LICENSE, README.md, requirements.txt, and solveitcore.py. The README is currently selected and displays the title "SOLVE-IT (a Systematic Objective-based Listing of Various Established digital Investigation Techniques)" and an introduction section.

SOLVE-IT (a Systematic Objective-based Listing of Various Established digital Investigation Techniques)

Introduction

Inspired by [MITRE ATT&CK](#), this repository contains a community project to map the objectives (tactics) and techniques that can be used as part of a digital forensic investigation.

SOLVE-IT structure: implementation

- The details are stored as JSON:

Name	
..	T1000.json
mitigations	T1001.json
techniques	T1002.json
weaknesses	T1003.json
carrier.json	T1004.json
dfrws.json	T1005.json
solve-it.json	T1006.json
	T1007.json
	T1008.json

```
1  {
2      "id": "T1002",
3      "name": "Disk imaging",
4      "description": "Copying of sectors from a storage media, typically LBA0 to LBAmx into an
5      "synonyms": [],
6      "details": "",
7      "subtechniques": [],
8      "examples": ["dcfldd", "FTK Imager", "Magnet ACQUIRE"],
9      "weaknesses": ["W1004", "W1006", "W1007", "W1013", "W1014", "W1015", "W1016", "W1136", "W1
10     "CASE_output_classes" : ["observable:Image"],
11     "references": ["Nikkel, B., 2016. Practical forensic imaging: securing digital evidence wi
12 }
```

SOLVE-IT structure: implementation

- There are scripts to convert the JSON into more usable formats (see later)

▼ reporting_scripts

generate_evaluation.py

generate_excel_from_kb.py

generate_stat_summary.py

generate_tsv_from_kb.py

```
[\\$ cd reporting_scripts
[\\$ python3 generate_excel_from_kb.py
INFO: Loaded 117 techniques.
INFO: Loaded 188 weaknesses.
INFO: Loaded 137 mitigations.
INFO: Building reverse indices for performance optimization...
INFO: Reverse indices built: 184 weakness->technique, 136 mitigation->weakness, 136 mitigation->technique
INFO: Loaded objective mapping 'solve-it.json' with 17 objectives.
Using configuration file: solve-it.json
Output will be to: output/solve-it.xlsx
Creating worksheets...
- added 'main' worksheet
- populated 'all techniques' worksheet
- populated 'all weaknesses' worksheet
- populated 'all techniques' worksheet
Updating 'main' worksheet with links to techniques...
- 'main' worksheet updated
Adding the individual techniques sheets...
- all individual techniques worksheets updated
[\\$ ls output
~$solve-it.xlsx solve-it.xlsx
\\$
```

- BUT pre-generated spreadsheets are available:

SOLVE-IT Release v0.2025-07

Latest

Compare



 chrishargreaves released this last week

· 15 commits to main since this release

 v0.2025-07

 fed891b



SOLVE-IT is a knowledge base of digital forensics techniques along with potential weaknesses and mitigations. It was launched in April at DFRWS EU 2025.

While the [GitHub repository](#) evolves every week we are going to issue some static releases at points throughout the year with some developments highlighted, along with accessible versions that can be downloaded for review and comment.

We are therefore happy to announce the first release: SOLVE-IT: v0.2025.07 (July 2025).

Since the DFRWS EU 2025 conference we have focused on some core improvements, documentation and accessibility, code refactoring, and testing.

Documentation and Accessibility

The [readme](#) has been updated along with a specific guidance document on [contributing](#).

The [issue tracker](#) has been updated with GitHub templates, making it easier to suggest new or updated techniques and weaknesses.

An [examples](#) repository has also been created with demonstrations of the use of SOLVE-IT to examine a sample disk imaging process.

Content updates

There are also many updates to the knowledge base content. We have now indexed:

- 117 techniques
- 188 weaknesses
- 137 mitigations

... and many more are recorded in the issue tracker waiting for further details and implementation.

Techniques that have been added include:

- Memory acquisition via cold boot attack,
- Determine connected devices,
- Visualisation of geolocation information, and
- Preserving reference data (inspired by [Spichiger & Adelstein \(2025\) talk at DFRWS EU](#)).

We are off to a good start getting the infrastructure in place, many good conversations starting, and the content is starting to grow.

A workshop is planned at [DFRWS USA 2025](#) in July so if you are interested in learning more about how to use SOLVE-IT or to contribute do [sign up for the workshop](#).

Assets3

 solve-it_v0.2025.07.xlsx	sha256:92b7dd2145f4949529f90c751...		150 KB	last week
 Source code (zip)				last week
 Source code (tar.gz)				last week

SOLVE-IT structure: implementation

- BUT pre-generated spreadsheets are available:
 - 1) manually generated
 - 2) automatically generated after each commit

chrishargreaves Update example_lab.json 5962723 · 1 hour ago 186 Commits		
.github	Update python-app.yml for new solve-it_library	1 hour ago
.repo_info	Update stats [skip ci]	1 hour ago
data	Update solve-it.json	last week
lab_config_examples	Update example_lab.json	1 hour ago
reporting_scripts	Add pre-computed reverse lookups.	15 hours ago
solve_it_library	Documentation updates.	14 hours ago
tests	Updating unit tests to account for recently added	15 hours ago
.gitignore	New library. Update generation scripts to use	18 hours ago
CONTRIBUTING.md	Update CONTRIBUTING.md	last month
LICENSE	Initial commit	7 months ago
README.md	New library. Update generation scripts to use	18 hours ago
requirements.txt	Documentation updates.	14 hours ago
solveitcore.py	removed extra new line	last month

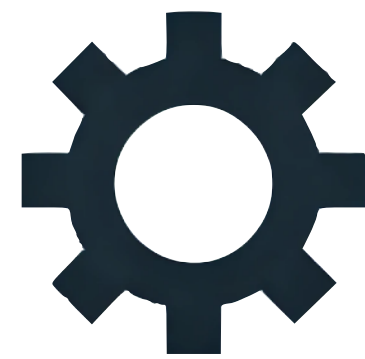
solve-it / .repo_info / Add file ...		
github-actions[bot] Update .repo_info [skip ci] b7e3249 · 4 minutes ago History		
Name	Last commit message	Last commit date
..		
CASE_mapping.txt	Update stats [skip ci]	last week
mitigations.txt	Update stats [skip ci]	yesterday
objectives.txt	Update stats [skip ci]	last week
solve-it-latest.xlsx	Update .repo_info [skip ci]	4 minutes ago
stats.txt	Update stats [skip ci]	last week
techniques.txt	Update stats [skip ci]	last week
techniques_long.txt	Update stats [skip ci]	last week
weaknesses.txt	Update stats [skip ci]	last week
weaknesses_long.txt	Update .repo_info [skip ci]	4 minutes ago

SOLVE-IT structure: design concepts



Objectives

The goal that one might wish to achieve in a digital forensic investigation, e.g. *acquire data* or *gain access*.



Techniques

How one might achieve an objective in digital forensics by performing an action, e.g. for the objective of 'acquire data', the technique 'disk imaging' could be used.



Weaknesses

These represent potential problems resulting from using a technique. They are classified according to the error categories in ASTM E3016-18.



Mitigations

Something that can be done to prevent a weakness from occurring, or to minimise its impact.

SOLVE-IT structure: relationships

```
1  {
2    "id": "T1002",
3    "name": "Disk imaging",
4    "description": "Copying of sectors from a storage media, typically LBA0 to LBAmx into an imaging format.
The could be from a traditional hard disk, SSD, USB stick, or data from an eMMC chip that has been desoldered
and placed in a reader. ",
5    "synonyms": [],
6    "details": "",
7    "subtechniques": [],
8    "examples": ["dcflddd", "FTK Imager", "Magnet ACQUIRE"],
9    "weaknesses": ["W1004", "W1006", "W1007", "W1013", "W1014", "W1015", "W1016", "W1136", "W1143"],
10   "CASE_output_classes" : ["observable:Image"],
11   "references": ["Nikkel, B., 2016. Practical forensic imaging: securing digital evidence with Linux tools.
No Starch Press, Chapter 6, "Forensic Image Acquisition"]
12 }
```

```
1  {
2    "id": "W1004",
3    "name": "Acquisition does not include all sectors from LBA0 to LBA max",
4    "INCOMP": "x",
5    "INAC-EX": "",
6    "INAC-AS": "",
7    "INAC-ALT": "",
8    "INAC-COR": "",
9    "MISINT": "",
10   "mitigations": ["M1003", "M1004"],
11   "references": []
12 }
```

```
1  {
2    "id": "W1006",
3    "name": "Acquistion does not include data in HPA",
4    "INCOMP": "x",
5    "INAC-EX": "",
6    "INAC-AS": "",
7    "INAC-ALT": "",
8    "INAC-COR": "",
9    "MISINT": "",
10   "mitigations": ["M1005"],
11   "references": ["Gupta, M.R., Hoeschele, M.D. and Rogers, M.K., 2006. Hidden disk areas: HPA and DCO.
International Journal of Digital Evidence, 5(1), pp.1-8."]
12 }
13
```

SOLVE-IT structure: relationships

ax into an imaging format.
p that has been desoldered

W1136", "W1143"],

evidence with Linux tools.

```
1 {
2   "id": "W1004",
3   "name": "Acquisition does not include all sectors from LBA0 to LBA max",
4   "INCOMP": "x",
5   "INAC-EX": "",
6   "INAC-AS": "",
7   "INAC-ALT": "",
8   "INAC-COR": "",
9   "MISINT": "",
10  "mitigations": ["M1003", "M1004"],
11  "references": []
12 }
```

```
1 {
2   "id": "W1006",
3   "name": "Acquistion does not include data in HPA",
4   "INCOMP": "x",
5   "INAC-EX": "",
6   "INAC-AS": "",
7   "INAC-ALT": "",
8   "INAC-COR": "",
9   "MISINT": "",
10  "mitigations": ["M1005"],
11  "references": ["Gupta, M.R., Hoeschele, M.D. and Rogers, M.K., 2006. Hidden disk areas: HPA and DCO.
International Journal of Digital Evidence, 5(1), pp.1-8."]
12 }
13
```

```
1 {
2   "id": "M1003",
3   "name": "Check image size corresponds with drive label",
4   "references": []
5 }
```

```
1 {
2   "id": "M1004",
3   "name": "Check hash of image matches hash of source device",
4   "references": []
5 }
```

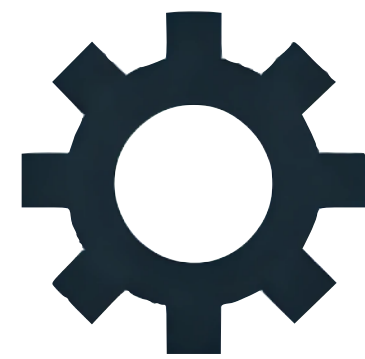
```
1 {
2   "id": "M1005",
3   "name": "Testing to ensure software and hardware setup detects HPAs",
4   "references": []
5 }
```

SOLVE-IT structure: design concepts



Objectives

The goal that one might wish to achieve in a digital forensic investigation, e.g. *acquire data* or *gain access*.



Techniques

How one might achieve an objective in digital forensics by performing an action, e.g. for the objective of 'acquire data', the technique 'disk imaging' could be used.



Weaknesses

These represent potential problems resulting from using a technique. They are classified according to the error categories in ASTM E3016-18.



Mitigations

Something that can be done to prevent a weakness from occurring, or to minimise its impact.



Objectives: Design

Survey	Preserve	Prioritise	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Locate Relevant Digital Artefacts	Extract Partition and System Information
Crime scene searching T1005	Place device in faraday environment T1010	Triage T1001	Disk imaging T1002	Key recovery from memory T1031	Disk image hash verification T1042	Privileged material protection T1046	Keyword searching T1049	Identify partitions T1059
Digital sniffer dogs T1006	Evidence bags T1011		Memory imaging T1003	Side channel T1032	Forensic image format decoding T1043	Hash matching (reduce) T1047	Hash matching (locate) T1050	Process file system structures T1060
SyncTriage-based approach T1007	Hardware write blockers T1012		Selective data acquisition T1004	Extraction of account details from an accessible device T1033	Mobile backup decoding T1044	Privacy protection via partial processing T1048	Fuzzy hash matching T1051	Non-allocated file recovery T1061
Profiling network traffic T1008	Software write blockers T1013		Privacy preserving selective extraction T1015	Brute force attack T1034	Decode standard archive format T1045		Timeline generation T1052	Decryption of encrypted file systems/volumes T1062
Locate cloud account identifiers T1009	Chain of custody documentation T1014		Live data collection T1016	Dictionary attack T1035	Decode data from image from unmanaged NAND T1102		Entity extraction T1053	Identify file types T1063
			Network packet capture T1017	Smudge attack T1036			Content review for relevant material T1054	File carving T1064
			Remote data collection T1018	Obtain password from suspect T1037			File system content inspection T1055	
			Mobile backup extraction	Rainbow tables			Entity connection	



Objectives: Design

“The goal that one might wish to achieve in a digital forensic investigation, e.g. acquire data or gain access.”

- Various process models that can be used for this
 - *Carrier* - acquire, analyse, present
 - *DFRWS/Palmer* - identification, preservation, collection, examination, analysis, presentation
 - *SOLVE-IT* - a new organisation, based on the need to categorise a large number of specific techniques



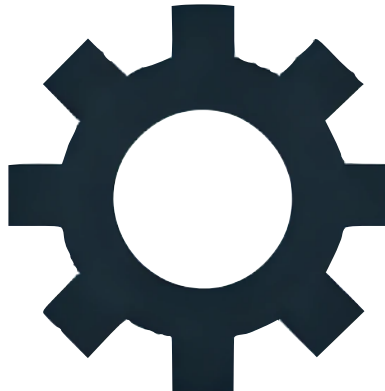
Objectives: Implementation

Name
..
mitigations
techniques
weaknesses
carrier.json
dfrws.json
solve-it.json

```
1  [  
2    {"name": "Survey",  
3     "description": "todo",  
4     "techniques": ["T1005", "T1006", "T1009", "T1008", "T1007"]  
5   },  
6  
7    {"name": "Preserve",  
8     "description": "todo",  
9     "techniques": ["T1014", "T1011", "T1010", "T1012", "T1013"]  
10  },  
11  
12   {"name": "Prioritise",  
13    "description": "todo",  
14    "techniques": ["T1001"]  
15  },  
16  
17   {"name": "Acquire",  
18    "description": "todo",  
19    "techniques": ["T1028", "T1023", "T1024", "T1029", "T1030",  
20                   "T1027", "T1002", "T1016", "T1003", "T1019",  
21                   "T1022", "T1020", "T1017", "T1015",  
22                   "T1018", "T1004", "T1026", "T1025", "T1104"]  
23  },  
24 ]
```

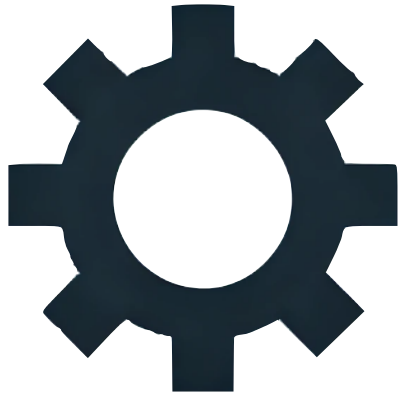
solve-it.json describes the primary organisation of the techniques, but...

... you can configure the SOLVE-IT tooling to use any different organisational structure needed.



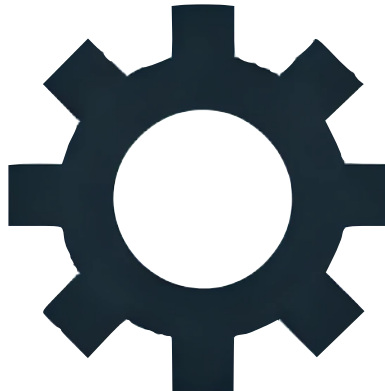
Techniques: Design

Survey	Preserve	Prioritise	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Locate Relevant Digital Artefacts	Extract Partition and System Information
Crime scene searching T1005	Place device in faraday environment T1010	Triage T1001	Disk imaging T1002	Key recovery from memory T1031	Disk image hash verification T1042	Privileged material protection T1046	Keyword searching T1049	Identify partitions T1059
Digital sniffer dogs T1006	Evidence bags T1011		Memory imaging T1003	Side channel T1032	Forensic image format decoding T1043	Hash matching (reduce) T1047	Hash matching (locate) T1050	Process file system structures T1060
SyncTriage-based approach T1007	Hardware write blockers T1012		Selective data acquisition T1004	Extraction of account details from an accessible device T1033	Mobile backup decoding T1044	Privacy protection via partial processing T1048	Fuzzy hash matching T1051	Non-allocated file recovery T1061
Profiling network traffic T1008	Software write blockers T1013		Privacy preserving selective extraction T1015	Brute force attack T1034	Decode standard archive format T1045		Timeline generation T1052	Decryption of encrypted file systems/volumes T1062
Locate cloud account identifiers T1009	Chain of custody documentation T1014		Live data collection T1016	Dictionary attack T1035	Decode data from image from unmanaged NAND T1102		Entity extraction T1053	Identify file types T1063
			Network packet capture T1017	Smudge attack T1036			Content review for relevant material T1054	File carving T1064
			Remote data collection T1018	Obtain password from suspect T1037			File system content inspection T1055	
			Mobile backup extraction	Rainbow tables			Entity connection	



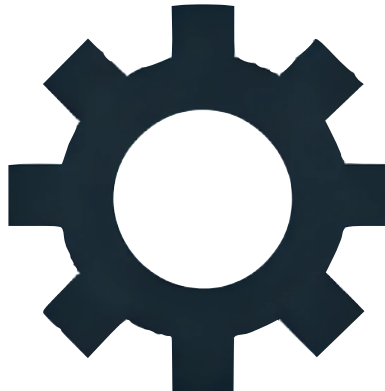
Techniques: Design

Survey	Preserve	Prioritise	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Locate Relevant Digital Artefacts	Extract Partition and System Information
Crime scene searching T1005	Place device in faraday environment T1010	Triage T1001	Disk imaging T1002	Key recovery from memory T1031	Disk image hash verification T1042	Privileged material protection T1046	Keyword searching T1049	Identify partitions T1059
Digital sniffer dogs T1006	Evidence bags T1011		Memory imaging T1003	Side channel T1032	Forensic image format decoding T1043	Hash matching (reduce) T1047	Hash matching (locate) T1050	Process file system structures T1060
SyncTriage-based approach T1007	Hardware write blockers T1012		Selective data acquisition T1004	Extraction of account details from an accessible device T1033	Mobile backup decoding T1044	Privacy protection via partial processing T1048	Fuzzy hash matching T1051	Non-allocated file recovery T1061
Profiling network traffic T1008	Software write blockers T1013		Privacy preserving selective extraction T1015	Brute force attack T1034	Decode standard archive format T1045		Timeline generation T1052	Decryption of encrypted file systems/volumes T1062
Locate cloud account identifiers T1009	Chain of custody documentation T1014		Live data collection T1016	Dictionary attack T1035	Decode data from image from unmanaged NAND T1102		Entity extraction T1053	Identify file types T1063
			Network packet capture T1017	Smudge attack T1036			Content review for relevant material T1054	File carving T1064
			Remote data collection T1018	Obtain password from suspect T1037			File system content inspection T1055	
			Mobile backup extraction	Rainbow tables			Entity connection	



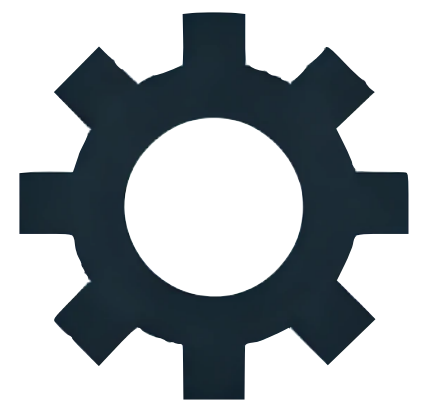
Techniques: Design

Survey	Preserve	Prioritise	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Locate Relevant Digital Artefacts	Extract Partition and System Information
Crime scene searching T1005	Place device in faraday environment T1010	Triage T1001	Disk imaging T1002	Key recovery from memory T1031	Disk image hash verification T1042	Privileged material protection T1046	Keyword searching T1049	Identify partitions T1059
Digital sniffer dogs T1006	Evidence bags T1011		Memory imaging T1003	Side channel T1032	Forensic image format decoding T1043	Hash matching (reduce) T1047	Hash matching (locate) T1050	Process file system structures T1060
SyncTriage-based approach T1007	Hardware write blockers T1012		Selective data acquisition T1004	Extraction of account details from an accessible device T1033	Mobile backup decoding T1044	Privacy protection via partial processing T1048	Fuzzy hash matching T1051	Non-allocated file recovery T1061
Profiling network traffic T1008	Software write blockers T1013		Privacy preserving selective extraction T1015	Brute force attack T1034	Decode standard archive format T1045		Timeline generation T1052	Decryption of encrypted file systems/volumes T1062
Locate cloud account identifiers T1009	Chain of custody documentation T1014		Live data collection T1016	Dictionary attack T1035	Decode data from image from unmanaged NAND T1102		Entity extraction T1053	Identify file types T1063
			Network packet capture T1017	Smudge attack T1036			Content review for relevant material T1054	File carving T1064
			Remote data collection T1018	Obtain password from suspect T1037			File system content inspection T1055	
			Mobile backup extraction	Rainbow tables			Entity connection	



Techniques: Design

Survey	Preserve	Prioritise	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Locate Relevant Digital Artefacts	Extract Partition and System Information
Crime scene searching T1005	Place device in faraday environment T1010	Triage T1001	Disk imaging T1002	Key recovery from memory T1031	Disk image hash verification T1042	Privileged material protection T1046	Keyword searching T1049	Identify partitions T1059
Digital sniffer dogs T1006	Evidence bags T1011		Memory imaging T1003	Side channel T1032	Forensic image format decoding T1043	Hash matching (reduce) T1047	Hash matching (locate) T1050	Process file system structures T1060
SyncTriage-based approach T1007	Hardware write blockers T1012		Selective data acquisition T1004	Extraction of account details from an accessible device T1033	Mobile backup decoding T1044	Privacy protection via partial processing T1048	Fuzzy hash matching T1051	Non-allocated file recovery T1061
Profiling network traffic T1008	Software write blockers T1013		Privacy preserving selective extraction T1015	Brute force attack T1034	Decode standard archive format T1045		Timeline generation T1052	Decryption of encrypted file systems/volumes T1062
Locate cloud account identifiers T1009	Chain of custody documentation T1014		Live data collection T1016	Dictionary attack T1035	Decode data from image from unmanaged NAND T1102		Entity extraction T1053	Identify file types T1063
			Network packet capture T1017	Smudge attack T1036			Content review for relevant material T1054	File carving T1064
			Remote data collection T1018	Obtain password from suspect T1037			File system content inspection T1055	
			Mobile backup extraction	Rainbow tables			Entity connection	



Techniques: Design

How one might achieve an objective in digital forensics by performing an action, e.g. for the objective of ‘acquire data’, the technique ‘disk imaging’ could be used.

- **id**: the technique’s ID, e.g. T1001;
- **name**: the name of the technique;
- **description**: A short description of what the technique involves;
- **synonyms**: any possible synonyms for the technique;
- **details**: further details beyond the short description;
- **sub-techniques**: some techniques may have sub-techniques, and can be listed here, referenced by technique ID;
- **examples**: examples related to the technique. These can be datasets that use the techniques, example cases that made use of the techniques either from published cases or synthetic ones, or specific tools that provide the technique;
- **weaknesses**: this field allows potential weaknesses associated with techniques to be referenced, pointing to indexed weaknesses within the knowledge base;
- **CASE_output_classes**: any potential CASE Ontology entities that allow the technique output to be represented;
- **references**: references can and should be included to support definitions and examples for the techniques.

Implementation: Techniques

Name
..
mitigations
techniques
weaknesses
carrier.json
dfrws.json
solve-it.json

T1000.json
T1001.json
T1002.json
T1003.json
T1004.json
T1005.json
T1006.json
T1007.json
T1008.json

```
1  {
2      "id": "T1002",
3      "name": "Disk imaging",
4      "description": "Copying of sectors from a storage media, typically LBA0 to LBA",
5      "synonyms": [],
6      "details": "",
7      "subtechniques": [],
8      "examples": ["dcfldd", "FTK Imager", "Magnet ACQUIRE"],
9      "weaknesses": ["W1004", "W1006", "W1007", "W1013", "W1014", "W1015", "W1016",
10     "CASE_output_classes" : ["observable:Image"],
11     "references": ["Nikkel, B., 2016. Practical forensic imaging: securing digital
12 }
```



Weaknesses: Design

	Survey	Preserve	Prioritise	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Locate Relevant Digital Artefacts	Extract Partition and System Information				
Crime	1	Technique name:		Disk imaging		back to main							
	2	Technique ID:		T1002									
	3	Category:		['Acquire']									
Digital				Copying of sectors from a storage media, typically LBA0 to LBAmx into an imaging format. The could be from a traditional hard disk, SSD, USB stick, or data from an eMMC chip that has been desoldered and placed in a reader.									
	4	Description:											
	5	Synonyms:		[]									
SyncTriag	6	Details:											
	7	Subtechniques:		[]									
	8	CASE output entities:		['observable:Image']									
Profilin	9	Examples:		['dcfldd', 'FTK Imager', 'Magnet ACQUIRE']									
	10												
	11	Potential weaknesses:											
Locate	12	Weakness ID:		Detail:		INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISINT	Mitigations	
	13	W1004		Acquisition does not include all sectors from LBA0 to LBA max		x						M1003,	
	14	W1006		Acquistion does not include data in HPA		x						M1005,	
	15	W1007		Acquistion does not include data in DCO		x						M1006,	
	16	W1013		Acquisition includes extra bytes			x					M1003,M1009,	
	17	W1014		Imaging process changes original data					x			M1007,M1008,	
	18	W1015		Powering on SSD results in sectors being wiped by TRIM operation		x			x	x			
	19	W1016		Data copied from sectors on source are stored incorrectly					x	x		M1009,	
	20	W1136		Not recovering data from a failed hard drive		x						M1089,	
	21	W1143		Acquisition method does not read remapped sectors e.g. G-Lists		x						M1102,	
	22												
	23	Mitigations:											

Weaknesses are presented when you look at a specific technique in the exported spreadsheet.

Weaknesses are presented when you look at a specific technique in the exported spreadsheet.



Weaknesses: Design

These represent potential problems resulting from using a technique. They are classified according to the error categories in ASTM E3016-18

- **id**: the weakness's ID (e.g. W1001);
- **name**: a short description of the weakness;
- **mitigations**: provides indexed references to any mitigations that could minimise or reduce the impact of individual weaknesses;
- **references**: These should be included to support definitions and examples, including to error-focused datasets demonstrating the weakness;
- **INCOMP**: weakness results in incompleteness;
- **INAC-EX**: weakness results in inaccuracy:existence;
- **INAC-AS**: weakness results in inaccuracy:association;
- **INAC-ALT**: weakness results in inaccuracy:alteration;
- **INAC-COR**: weakness results in inaccuracy:corruption;
- **MISINT**: weakness results in potential misinterpretation;



Weaknesses: Implementation

Name	
	W1001.json
..	W1002.json
mitigations	W1003.json
techniques	W1004.json
weaknesses	W1005.json
carrier.json	W1006.json
dfrws.json	W1007.json
solve-it.json	W1008.json
	W1009.json
	W1010.json

```
1  {
2      "id": "W1004",
3      "name": "Acquisition does not include all sectors from LBA0 to LBA max",
4      "INCOMP": "x",
5      "INAC-EX": "",
6      "INAC-AS": "",
7      "INAC-ALT": "",
8      "INAC-COR": "",
9      "MISINT": "",
10     "mitigations": ["M1003", "M1004"],
11     "references": []
12 }
```



Mitigations: Design

Survey	Preserve	Prioritise	Acquire	Gain Access	Process Storage Format	Perform Data Reduction	Locate Relevant Digital Artefacts	Extract Partition and System Information	
Crime scene snapshot T1001	1	Technique name:	Disk imaging		back to main				
	2	Technique ID:	T1002						
	3	Category:	['Acquire']						
Digital sniffing T1002	4	Description:	Copying of sectors from a storage media, typically LBA0 to LBAmx into an imaging format. The could be from a traditional hard disk, SSD, USB stick, or data from an eMMC chip that has been desoldered and placed in a reader.						
SyncTriage-based T1003	5	Synonyms:	[]						
	6	Details:							
	7	Subtechniques:	[]						
Profiling network T1004	8	CASE output entities:	['observable:Image']						
	9	Examples:	['dcfldd', 'FTK Imager', 'Magnet ACQUIRE']						
	10								
Locate cloud storage T1005	11	Potential weaknesses:							
	12	Weakness ID:	Detail:	INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISINT
	13	W1004	Acquisition does not include all sectors from LBA0 to LBA max	x					
	14	W1006	Acquisition does not include data in HPA	x					
	15	W1007	Acquisition does not include data in DCO	x					
	16	W1013	Acquisition includes extra bytes		x				
	17	W1014	Imaging process changes original data				x		
	18	W1015	Powering on SSD results in sectors being wiped by TRIM operation	x			x	x	
	19	W1016	Data copied from sectors on source are stored incorrectly				x	x	
	20	W1136	Not recovering data from a failed hard drive	x					
	21	W1143	Acquisition method does not read remapped sectors e.g. G-Lists	x					
	22								
	23	Mitigations:							
	24	M1003	Check image size corresponds with drive label						

Mitigations are mapped to, and visible within specific weaknesses...

M1003,
M1005,
M1006,
M1003,M1009,
M1007,M1008,

M1009,
M1089,
M1102,

Mitigations are mapped to, and visible within specific weaknesses...

Mitigations
M1003,
M1005,
M1006,
M1003,M1009,
M1007,M1008,

M1009,
M1089,
M1102,



1	Technique name:	Disk imaging	back to main						
2	Technique ID:	T1002							
3	Category:	['Acquire']							
4	Description:	Copying of sectors from a storage media, typically LBA0 to LBAmix into an imaging format. The could be from a traditional hard disk, SSD, USB stick, or data from an eMMC chip that has been desoldered and placed in a reader.							
5	Synonyms:	[]							
6	Details:								
7	Subtechniques:	[]							
8	CASE output entities:	['observable:Image']							
9	Examples:	['dcfldd', 'FTK Imager', 'Magnet ACQUIRE']							
10									
11	Potential weaknesses:								
12	Weakness ID:	Detail:	INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISINT	Mitigations
13	W1004	Acquisition does not include all sectors from LBA0 to LBA max	x						M1003,
14	W1006	Acquistion does not include data in HPA	x						M1005,
15	W1007	Acquistion does not include data in DCO	x						M1006,
16	W1013	Acquisition includes extra bytes		x					M1003,M1009,
17	W1014	Imaging process changes original data				x			M1007,M1008,
18	W1015	Powering on SSD results in sectors being wiped by TRIM operation	x			x	x		
19	W1016	Data copied from sectors on source are stored incorrectly				x	x		M1009,
20	W1136	Not recovering data from a failed hard drive	x						M1089,
21	W1143	Acquisition method does not read remapped sectors e.g. G-Lists	x						M1102,
22									
23	Mitigations:								
24	M1003	Check image size corresponds with drive label							
25	M1005	Testing to ensure software and hardware setup detects HPAs							
26	M1006	Testing to ensure software and hardware setup detects DCOs							
27	M1007	Use hardware write blocker (T1012)							
28	M1008	Use software write blocker (T1013)							
29	M1009	Check hash of image matches hash of source material							
30	M1089	Attempt physical disk repair							
31	M1102	Apply techniques to read remapped sectors							
32									

... with the detail provided below.

... with the detail provided below.



Mitigations: Design

Something that can be done to prevent a weakness from occurring, or to minimise its impact

- **id**: the mitigation's ID (e.g. M1001);
- **name**: a short description of the mitigation;
- **details**: A longer description for the mitigation;
- **technique**: an optional index to a related technique. This can be used when a mitigation is sufficiently complex to be considered a technique in its own right;
- **references**: these should be included to support the description of the mitigation.



Mitigations: Implementation

Go to file

t

M1037.json

M1038.json

M1039.json

M1040.json

M1041.json

M1042.json

Code

Blame

5 lines (5 loc) · 145 Bytes

Code 55% faster with GitHub Copilot

```
1  {
2      "id": "M1038",
3      "name": "Word list selected such that a practically reviewable number of results are returned",
4      "references" : []
5  }
```

This mitigation is referenced from W1059 (excessive keyword results returned)

```
1  {
2      "id": "W1059",
3      "name": "Excessive results returned such that careful review of all results is impractical",
4      "INCOMP": "X",
5      "INAC-EX": "",
6      "INAC-AS": "",
7      "INAC-ALT": "",
8      "INAC-COR": "",
9      "MISINT": "",
10     "mitigations": ["M1032", "M1033", "M1038"],
11     "references": []
12 }
```

SOLVE-IT report generation

SOLVE-IT report generation

- There are scripts to convert the JSON into more usable formats

▼ reporting_scripts

generate_evaluation.py

generate_excel_from_kb.py

generate_stat_summary.py

generate_tsv_from_kb.py

```
[\\$ cd reporting_scripts
[\\$ python3 generate_excel_from_kb.py
INFO: Loaded 117 techniques.
INFO: Loaded 188 weaknesses.
INFO: Loaded 137 mitigations.
INFO: Building reverse indices for performance optimization...
INFO: Reverse indices built: 184 weakness->technique, 136 mitigation->weakness, 136 mitigation->technique
INFO: Loaded objective mapping 'solve-it.json' with 17 objectives.
Using configuration file: solve-it.json
Output will be to: output/solve-it.xlsx
Creating worksheets...
- added 'main' worksheet
- populated 'all techniques' worksheet
- populated 'all weaknesses' worksheet
- populated 'all techniques' worksheet
Updating 'main' worksheet with links to techniques...
- 'main' worksheet updated
Adding the individual techniques sheets...
- all individual techniques worksheets updated
[\\$ ls output
~$solve-it.xlsx solve-it.xlsx
\\$
```

SOLVE-IT report generation

- Pre-generated versions are in the 'Releases' section, and autogenerated for each commit (in /.repo_info).

 Public

Unwatch 7

Fork 5

Starred 39

main 9 Branches 2 Tags

Go to file

Add file

Code

 chrishargreaves Update example_lab.json 5962723 · 1 hour ago 186 Commits

.github	Update python-app.yml for new solve-it_library	1 hour ago
.repo_info	Update stats [skip ci]	1 hour ago
data	Update solve-it.json	last week
lab_config_examples	Update example_lab.json	1 hour ago
reporting_scripts	Add pre-computed reverse lookups.	15 hours ago
solve_it_library	Documentation updates.	13 hours ago
tests	Updating unit tests to account for recently added	15 hours ago
.gitignore	New library. Update generation scripts to use	18 hours ago
CONTRIBUTING.md	Update CONTRIBUTING.md	last month
LICENSE	Initial commit	7 months ago
README.md	New library. Update generation scripts to use	18 hours ago
requirements.txt	Documentation updates.	13 hours ago
solveitcore.py	removed extra new line	last month

About

The SOLVE-IT knowledge base for digital forensics

Readme

MIT license

Activity

39 stars

7 watching

5 forks

Report repository

Releases 2

 SOLVE-IT Release v0.2025-07 Latest

last week

+ 1 release

Packages

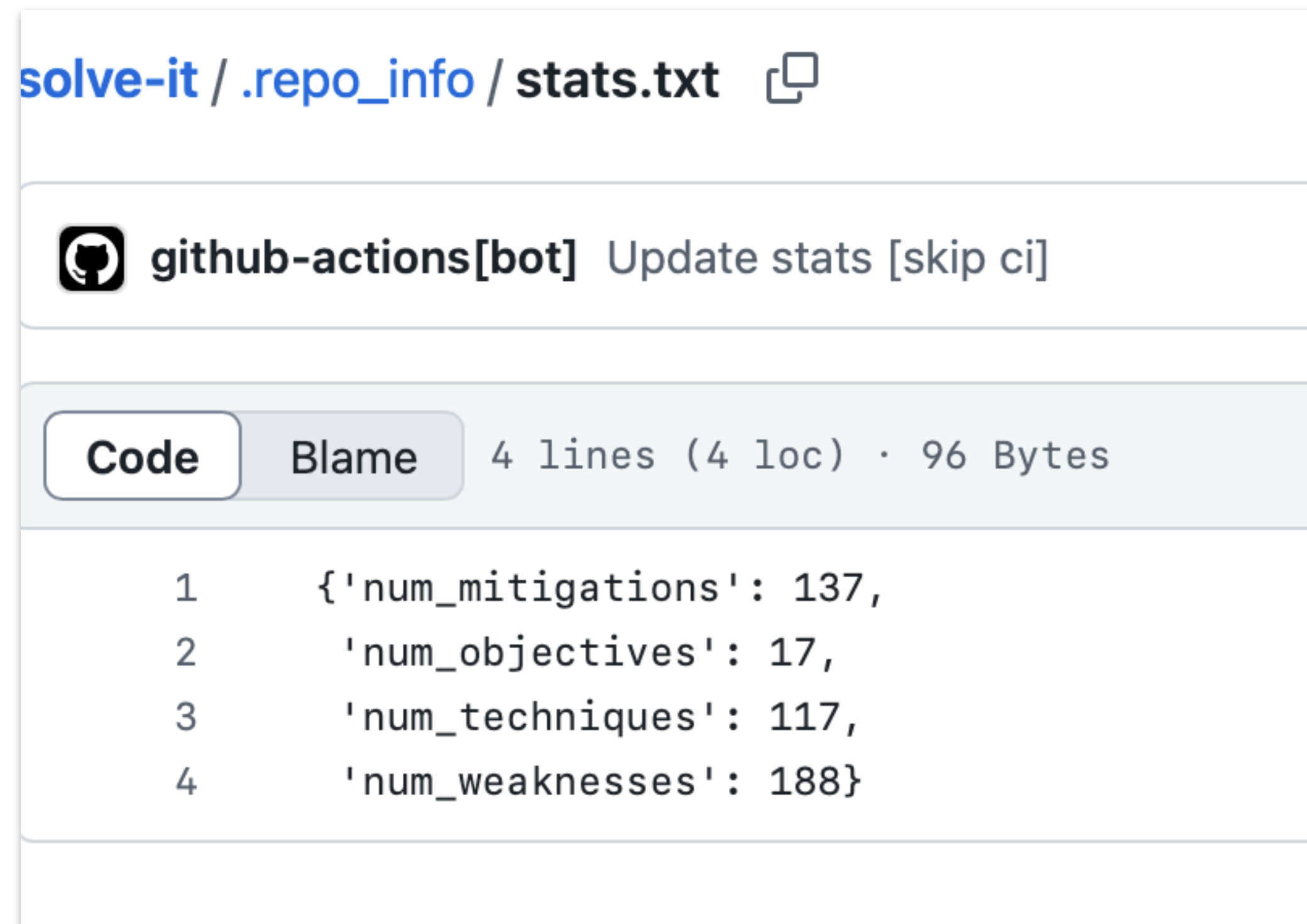
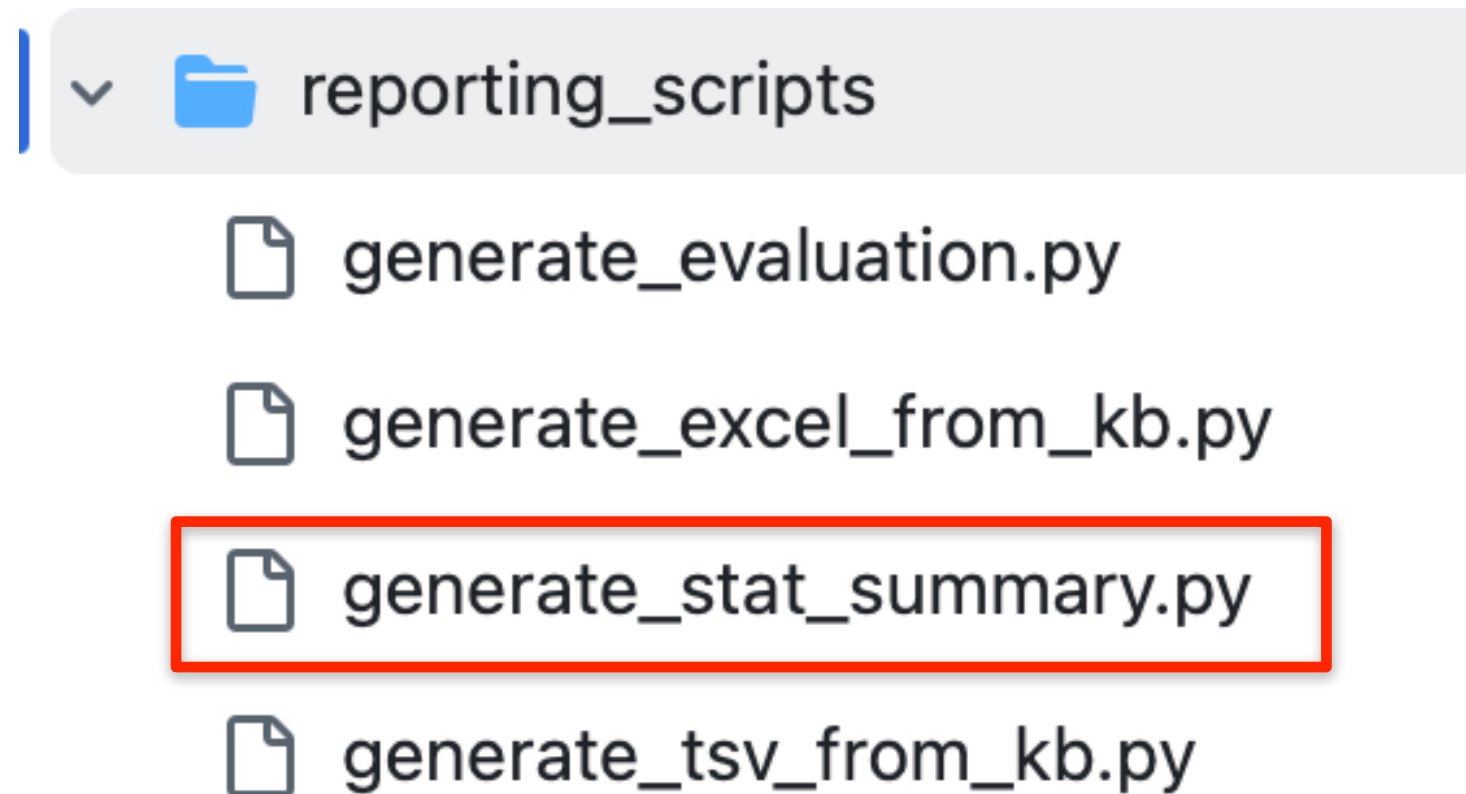
No packages published

[Publish your first package](#)

Contributors 6

SOLVE-IT report generation

- There are scripts to convert the JSON into more usable formats



SOLVE-IT report generation

- There are scripts to convert the JSON into more usable formats

▼  reporting_scripts

 generate_evaluation.py

 generate_excel_from_kb.py

 generate_stat_summary.py

 generate_tsv_from_kb.py

```
[\$ python3 generate_tsv_from_kb.py
usage: generate_tsv_from_kb.py [-h] [--objectives] [--techniques] [--weaknesses] [--mitigations] [--case] [--long]

Generate TSV output for parts of the SOLVE-IT knowledge base

options:
  -h, --help            show this help message and exit
  --objectives, -o      Print the objectives from SOLVE-IT in TSV format
  --techniques, -t      Print the techniques from SOLVE-IT in TSV format
  --weaknesses, -w      Print the weaknesses from SOLVE-IT in TSV format
  --mitigations, -m     Print the mitigations from SOLVE-IT in TSV format
  --case, -c            Print the mapping of techniques to CASE ontology
  --long, -l            Print extended fields other than ID and name
\$
```

▼  .repo_info

 CASE_mapping.txt

 mitigations.txt

 objectives.txt

 stats.txt

 techniques.txt

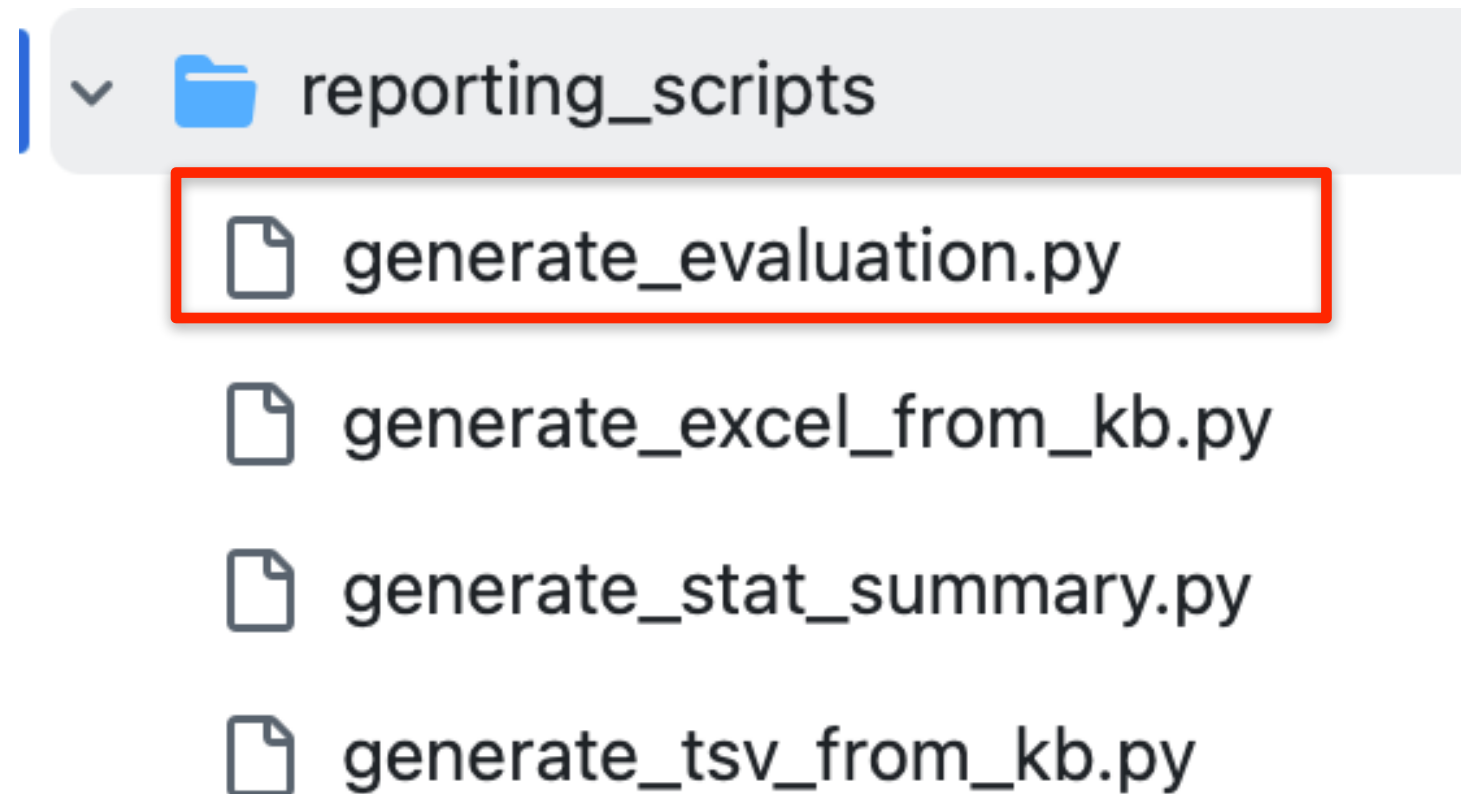
 techniques_long.txt

 weaknesses.txt

 weaknesses_long.txt

SOLVE-IT report generation

- There are scripts to convert the JSON into more usable formats



```
[\\$ cd reporting_scripts
[\\$ python3 generate_evaluation.py
INFO: Loaded 117 techniques.
INFO: Loaded 188 weaknesses.
INFO: Loaded 137 mitigations.
INFO: Building reverse indices for performance optimization...
INFO: Reverse indices built: 184 weakness->technique, 136 mitigation->weakness, 136 mitigation->technique
INFO: Loaded objective mapping 'solve-it.json' with 17 objectives.
Max mitigations: 10
['T1000', 'T1001', 'T1002', 'T1003', 'T1004', 'T1005', 'T1006', 'T1007', 'T1008', 'T1009', 'T1010', 'T1011', 'T1012', 'T1013', 'T1014', 'T1015', 'T1016', 'T1017', 'T1018', 'T1019', 'T1020', 'T1021', 'T1022', 'T1023', 'T1024', 'T1025', 'T1026', 'T1027', 'T1028', 'T1029', 'T1030', 'T1031', 'T1032', 'T1033', 'T1034', 'T1035', 'T1036', 'T1037', 'T1038', 'T1039', 'T1040', 'T1041', 'T1042', 'T1043', 'T1044', 'T1045', 'T1046', 'T1047', 'T1048', 'T1049', 'T1050', 'T1051', 'T1052', 'T1053', 'T1054', 'T1055', 'T1056', 'T1057', 'T1058', 'T1059', 'T1060', 'T1061', 'T1062', 'T1063', 'T1064', 'T1065', 'T1066', 'T1067', 'T1068', 'T1069', 'T1070', 'T1071', 'T1072', 'T1073', 'T1074', 'T1075', 'T1076', 'T1077', 'T1078', 'T1079', 'T1080', 'T1081', 'T1082', 'T1083', 'T1084', 'T1085', 'T1086', 'T1087', 'T1088', 'T1089', 'T1090', 'T1091', 'T1092', 'T1093', 'T1094', 'T1095', 'T1096', 'T1097', 'T1098', 'T1099', 'T1100', 'T1101', 'T1102', 'T1103', 'T1104', 'T1105', 'T1106', 'T1107', 'T1108', 'T1109', 'T1110', 'T1111', 'T1112', 'T1113', 'T1114', 'T1115', 'T1116', 'T1117', 'T1118']
Evaluation workbook successfully generated at: output/solve-it_evaluation_workbook.xlsx
\\$
```


SOLVE-IT report generation

SOLVE-IT-DF

Overview

Repositories4

Projects

Packages

Stars

Type / to search



SOLVE-IT

SOLVE-IT-DF

Follow

7 followers · 0 following

Popular repositories

solve-it

The SOLVE-IT knowledge base for digital forensics

Python 39 5

solve-it-examples

Practical examples using the SOLVE-IT knowledge base.

solve-it-applications-ai-review

An application of SOLVE-IT to review AI applicability to DF techniques

TeX 2 1

solve-it-education

Material that can be used for teaching using SOLVE-IT

24 contributions in the last year

	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
Mon													
Wed													
Fri													

Learn how we count contributions

Less More

2025

2024

SOLVE-IT report generation

SOLVE-IT-DF / solve-it-examples

<> Code

Issues

Pull requests

Actions

Projects

Security

Insights

🔍

Type / to search

+

🔍

solve-it-examples

Public

👁 Unwatch

2

🍴 Fork

0

★ Star

0

🔗 main

🔗 1 Branch

🏷 0 Tags

🔍

Go to file

t

Add file

<> Code

👤

chrishargreaves

Update README.md

7c55e6d · 3 months ago

🕒 14 Commits

📁 dependencies_in_digital_forensic_tools	Added placeholders for other examples	3 months ago
📁 forensic_workflow_example_forensic_imaging	Fixed link to XLXS	3 months ago
📁 mapping_of_techniques_a_drone_forensic_...	Added placeholders for other examples	3 months ago
📁 subtleties_in_mitigation_approaches_a_file...	Added placeholders for other examples	3 months ago
📄 LICENSE	Initial commit	3 months ago
📄 README.md	Update README.md	3 months ago

📖 README

📄 MIT license

✎

☰

A series of examples making use of the SOLVE-IT knowledge base

About

Practical examples using the SOLVE-IT knowledge base.

📖 Readme

📄 MIT license

📈 Activity

★ 0 stars

👁 2 watching

🍴 0 forks

Report repository

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

SOLVE-IT report generation

Forensic workflow example: forensic imaging

Introduction

This example shows how SOLVE-IT can be used to examine a forensic workflow and consider the weaknesses in the process and evaluate the mitigations in place.

An example process

For this example we can take the process required to acquire data from a hard disk.

Referred to broadly as 'disk imaging', when mapped as a *process* against SOLVE-IT techniques, there are several that are used. For example:

- T1012: Hardware write blocker[\[1\]](#)
- T1002: Disk imaging
- T1025: Writing to a forensic image
- T1042: Disk image hash verification

SOLVE-IT report generation

Using SOLVE-IT

The utility python script `generate_evaluation.py` can be used here.

Running the script with those technique IDs as parameters, compiles an Excel spreadsheet that extracts the weaknesses and available mitigations into a reviewable form.

```
python generate_evaluation.py T1012 T1002 T1025 T1042
```



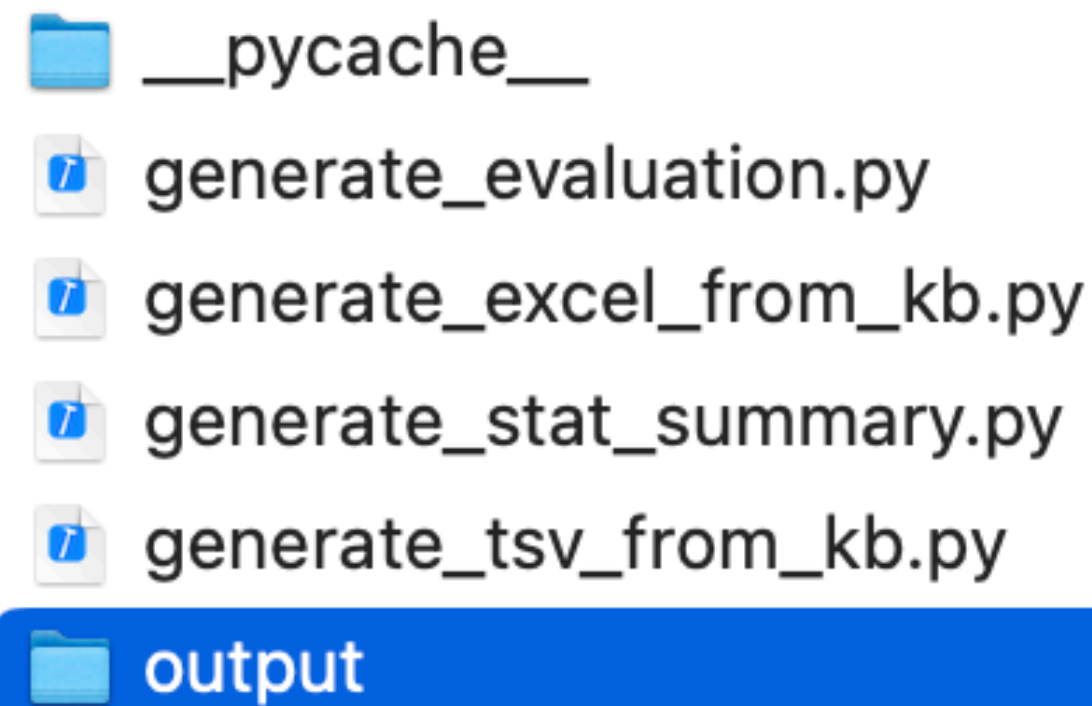
This output will highlight that, for example, under the technique T1012: Hardware write blocker, there are currently three potential weaknesses

- Hardware write blocker fails to prevent modifications to the attached device.
- Hardware write blocker hides the existence of an HPA.
- Hardware write blocker hides the existence of an DCO.

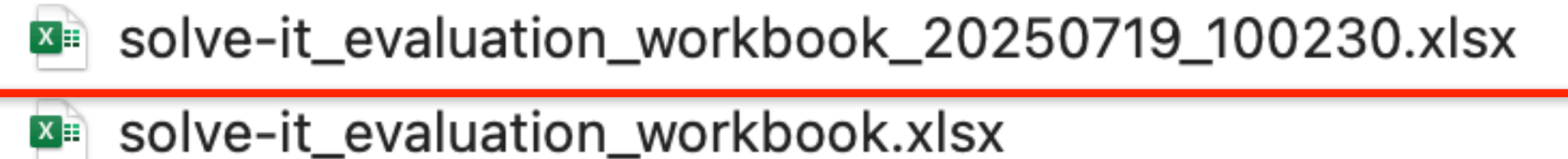
The figure below shows a section of the example output mapping the available mitigations against those potential weaknesses.

SOLVE-IT report generation

```
[\$ cd reporting_scripts
[\$ python3 generate_evaluation.py T1012 T1002 T1025 T1042
Output file already exists. Using new filename: output/solve-it_evaluation_workbook_20250719_100230.xlsx
INFO: Loaded 117 techniques.
INFO: Loaded 188 weaknesses.
INFO: Loaded 137 mitigations.
INFO: Building reverse indices for performance optimization...
INFO: Reverse indices built: 184 weakness->technique, 136 mitigation->weakness, 136 mitigation->technique
INFO: Loaded objective mapping 'solve-it.json' with 17 objectives.
Max mitigations: 10
['T1012', 'T1002', 'T1025', 'T1042']
Evaluation workbook successfully generated at: output/solve-it_evaluation_workbook_20250719_100230.xlsx
\$
```



- __pycache__
- generate_evaluation.py
- generate_excel_from_kb.py
- generate_stat_summary.py
- generate_tsv_from_kb.py
- output



- solve-it_evaluation_workbook_20250719_100230.xlsx
- solve-it_evaluation_workbook.xlsx

SOLVE-IT report generation

[illegible]

T1012	Hardware write blockers	INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISIN
W1118	Hardware write blocker fails to prevent modifications to the attached device.					X	
W1119	Hardware write blocker hides the existence of an HPA.	X					
W1120	Hardware write blocker hides the existence of an DCO.	X					
T1002	Disk imaging	INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISIN
W1004	Acquisition does not include all sectors from LBA0 to LBA max	x					
W1006	Acquistion does not include data in HPA	x					
W1007	Acquistion does not include data in DCO	x					
W1013	Acquisition includes extra bytes		x				
W1014	Imaging process changes original data				x		
W1015	Powering on SSD results in sectors being wiped by TRIM operation	x			x	x	
W1016	Data copied from sectors on source device are stored incorrectly				x	x	
W1136	Not recovering data from a failed hard drive	x					
W1143	Acquisition method does not read remapped sectors e.g. G-Lists	x					
T1025	Writing data to a forensic image format	INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISIN
W1043	Data is written to forensic format that does not preserve the original raw data				X		
T1042	Disk image hash verification	INCOMP	INAC-EX	INAC-AS	INAC-ALT	INAC-COR	MISIN
W1042	Disk image was tampered with, but manipulated to have a collision with original hash					x	
W1124	Failure to compute hash correctly: this could result in a message indicating corrupt evidence, thus stopping or delaying further investigation			X			
W1125	Failure to validate hash properly: this could allow errors from earlier to propagate e.g. incorrect sectors					X	
W1126	Failure to validate hash properly allowing an incomplete disk image to present as validated	X					
W1127	Failure to validate metadata: this could allow details such as acquisition date to be changed					X	
W1128	Image replaced with tampered version along with updated stored hash					X	

[illegible]

M1071 Testing write blocker multiple times to ensure it is not disabled.	M1072 Regular checks for hardware write blocker firmware updates.	M1073 Subscription to notifications from write blocker vendor for firmware updates or identified problems.	M1005 Testing to ensure software and hardware setup detects HPAs	M1006 Testing to ensure software and hardware setup detects DCOs																					
	-	-								0	0	3	0	3	0/3	x									
			-								0	0	1	0	1	0/1	x								
				-								0	0	1	0	1	0/1	x							
M1003 Image size matches label	M1004 Check hash of image matches hash of source device	M1005 Testing to ensure software and hardware setup detects HPAs	M1006 Testing to ensure software and hardware setup detects DCOs	M1007 Use hardware write blocker	M1008 Use software write blocker	M1089 Attempt physical disk repair	M1102 Apply techniques to read remapped sectors																		
	Y									2	0	0	0	2	2/2										
		Y								1	0	0	0	1	1/1		Standard imaging setup tested with HPA disk TTP-HPA-01								
			Y								1	0	0	0	1	1/1		Standard imaging setup tested with DCO disk TTP-DCO-01							
	-									0	0	2	0	2	0/2	x									
				Y	NA								1	0	0	1	1/1		Software write blockers are not used						
		-								0	0	0	0	0	0/0										
						N								0	0	1	0	1	0/1	x					
							N								0	1	0	0	1	0/1	x	Physical repair not undertaken at this lab			
M1009 Hash of image matches source file																									
										0	0	1	0	1	0/1	x									
M1021 Verify disk integrity using multiple hash algorithms e.g. SHA1 (NIST SP 800-57 Rev 1, 2016)	M1022 Restrict access to stored disk images	M1023 Ensure and check logs of access to stored disk images	M1070 Ensure hash algorithm(s) used are resistant to collisions through data manipulation	M1075 Testing programme to validate hashes of data in images is calculated correctly	M1085 Use of multiple tools to verify disk image hash	M1076 Testing programme to validate hashes of metadata in images is calculated correctly	M1074 Validate image hash against one stored externally to the image in a trusted location.																		
	-	-	-								0	0	4	0	4	0/4	x								
				-	-								0	0	2	0	2	0/2	x						
				-	-								0	0	2	0	2	0/2	x						
				-	-								0	0	2	0	2	0/2	x						
						-								0	0	1	0	1	0/1	x					
	-	-								-								0	0	3	0	3	0/3	x	

SOLVE-IT report generation

- It is also possible to use 'lab configurations' to auto-populate values during worksheet generation.

```
{
  "Lab config notes": "This is a demo to be used with: generate_evaluation.py T1002 --lab_config ../lab_config_examples/example_lab.json ",
  "T1002:Disk imaging": {
    "W1006": {
      "M1005": {
        "status": "Y",
        "notes": "Standard imaging setup tested with HPA disk TTP-HPA-01"
      }
    },
    "W1007": {
      "M1006": {
        "status": "Y",
        "notes": "Standard imaging setup tested with DCO disk TTP-DCO-01"
      }
    },
    "W1014": {
      "M1007": {
        "status": "Y",
        "notes": "Hardware write blockers are always used according to lab policy LAB_TK-421"
      },
      "M1008": {
        "status": "NA",
        "notes": "Software write blockers are not used"
      }
    },
    "W1136": {
      "M1089": {
        "status": "N",
        "notes": "Physical repair not undertaken at this lab"
      }
    },
    "W1143": {
      "M1102": {
        "status": "N",
        "notes": "Tooling for recovery of remapped sectors not available at this lab"
      }
    }
  }
}
```

solve-it/lab_config_examples/example_lab.json

SOLVE-IT report generation

```
{
  "Lab config notes": "This is a demo to be used with: generate_evaluation.py T1002 --lab_config ../lab_config_examples/example_lab.json ",
  "T1002:Disk imaging": {
    "W1006": {
      "M1005": {
        "status": "Y",
        "notes": "Standard imaging setup tested with HPA disk TTP-HPA-01"
      }
    },
    "W1007": {
      "M1006": {
        "status": "Y",
        "notes": "Standard imaging setup tested with DCO disk TTP-DCO-01"
      }
    },
    "W1014": {
      "M1007": {
        "status": "Y",
        "notes": "Hardware write blockers are always used according to lab policy LAB_TK-421"
      },
      "M1008": {
        "status": "NA",
        "notes": "Software write blockers are not used"
      }
    },
    "W1015": {
      "M1009": {
        "status": "Y",
        "notes": "Standard imaging setup tested with HPA disk TTP-HPA-02"
      }
    }
  }
}
```

SOLVE-IT report generation

```
[\\$ ls lab_config_examples
example_lab_tool_test.json      example_lab.json
[\\$ cd reporting_scripts
[\\$ python3 generate_evaluation.py T1002 --lab_config ../lab_config_examples/example_lab.json
INFO: Loaded 117 techniques.
INFO: Loaded 188 weaknesses.
INFO: Loaded 137 mitigations.
INFO: Building reverse indices for performance optimization...
INFO: Reverse indices built: 184 weakness->technique, 136 mitigation->weakness, 136 mitigation->technique
INFO: Loaded objective mapping 'solve-it.json' with 17 objectives.
INFO: Lab configuration loaded from: ../lab_config_examples/example_lab.json
Max mitigations: 10
['T1002']
INFO: Techniques in lab config: ['T1002:Disk imaging']
INFO: Weaknesses in lab config: ['W1006', 'W1007', 'W1014', 'W1136', 'W1143']
INFO: Mitigations in lab config: ['M1005', 'M1006', 'M1007', 'M1008', 'M1089', 'M1102']
Evaluation workbook successfully generated at: output/solve-it_evaluation_workbook.xlsx
\\$ █
```

SOLVE-IT report generation

[illegible]

SOLVE-IT report generation

Examples available here:

https://github.com/SOLVE-IT-DF/solve-it-examples/blob/main/forensic_workflow_example_forensic_imaging/README.md

[solve-it-examples](#) / [forensic_workflow_example_forensic_imaging](#) / [README.md](#) 

 **chrishargreaves** Update README.md with lab config example 8a19a1d · 1 minute ago  History

Preview Code Blame

138 lines (94 loc) · 5.79 KB

 Raw     

Forensic workflow example: forensic Imaging

Introduction

This example shows how SOLVE-IT can be used to examine a forensic workflow and consider the weaknesses in the process and evaluate the mitigations in place.

An example process

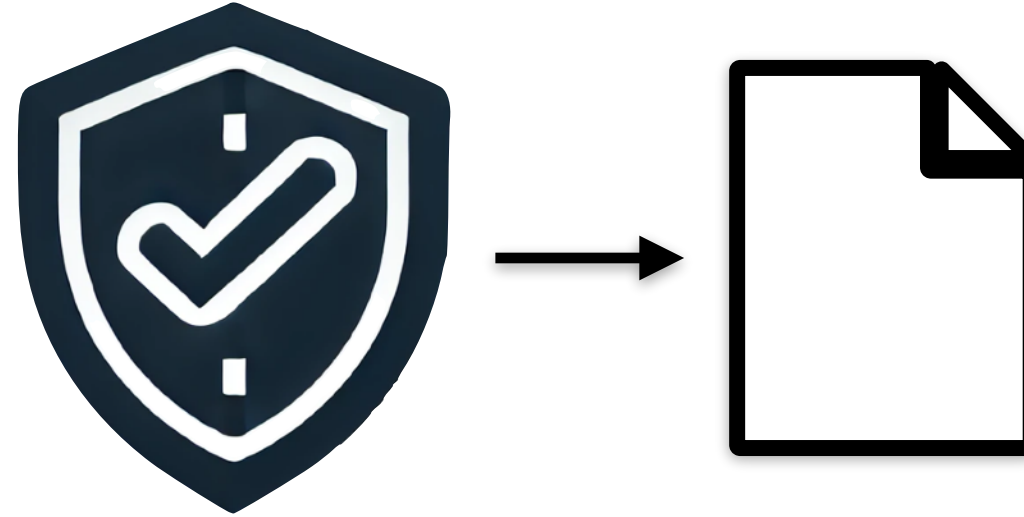
For this example we can take the process required to acquire data from a hard disk.

Referred to broadly as 'disk imaging', when mapped as a *process* against SOLVE-IT techniques, there are several that are used. For example:

SOLVE-IT context: other applications



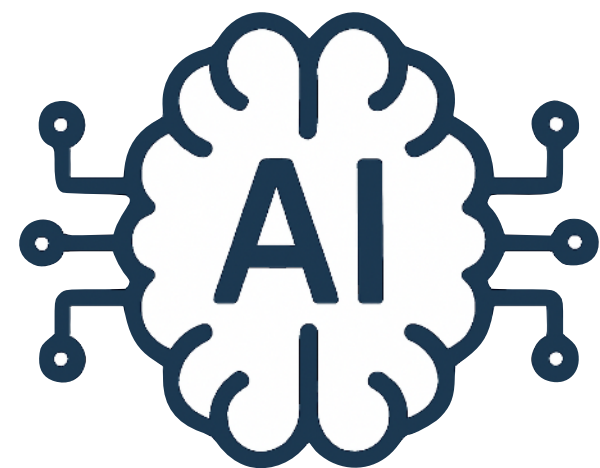
Scoping error
focused
datasets



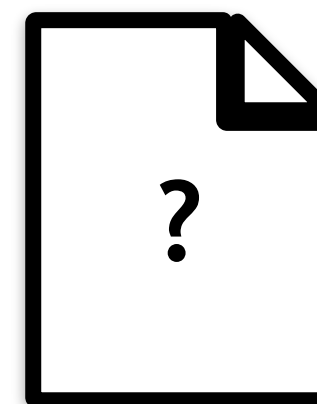
Highlighting mitigations
that exist for a weakness
in a technique



Identifying weaknesses
in a case, SOP/process,
or tool workflow



Structured
consideration of AI
applications



Academic research gaps
(research directions)

...

SOLVE-IT report generation

- Summary
 - There are multiple reporting scripts available in the repository.
 - Generate stats
 - Generate TSV versions of techniques, weaknesses, mitigations for quick scripts and basic LLM use.
 - Generate Excel version of the knowledge base
 - Generate a 'SOLVE-IT Evaluation Workbook' to systematically review the mitigations that are in place for techniques that have been used and identify unmitigated weaknesses.

SOLVE-IT
tool
integration?

SOLVE-IT tool integration?

ALEAPP Demo:

Integrating a SOLVE-IT-based logger

SOLVE-IT:
contributing

SOLVE-IT: contributions needed!

- **SOLVE-IT content contributor**

- Writes content for the SOLVE-IT knowledge base, either techniques, weaknesses, or mitigations.

- **SOLVE-IT knowledge base editor**

- Creates JSON data for integration directly into the knowledge base.

- **SOLVE-IT technical reviewer**

- Reviews the submissions from the content creators and implementers to check details before they are merged into the repository.

- **SOLVE-IT developer**

- Develops new scripts, refines old ones, bug fixes, GitHub automation work, UI development etc.

SOLVE-IT: contributing

Contributing to SOLVE-IT

THIS DOCUMENT IS UNDER DEVELOPMENT

Introduction

There are multiple ways you can contribute to SOLVE-IT.

Content that can be contributed

There are many areas in which content can be contributed:

- **Techniques** - how one might achieve an objective in digital forensics by performing an action, e.g. for the objective of 'acquire data', the technique 'create disk image' could be used.
- **Weaknesses** - these represent potential problems resulting from using a technique.
- **Mitigations** - something that can be done to prevent a weakness from occurring, or to minimise its impact.
- **Examples** - examples of cases where a technique is used, datasets that are relevant to the technique, tools that are able to perform the technique.
- **References** - references to support the information within techniques, weaknesses and mitigations. We are working on a [bibtex implementation for references](#).

Please also see the [Review of AI applicability in digital forensics using SOLVE-IT](#) repo, which invites bibtex entries that document the use of AI for assisting with digital forensic techniques.

<https://github.com/SOLVE-IT-DF/solve-it/blob/main/CONTRIBUTING.md>

SOLVE-IT: contributing

Methods to contribute

We have two main ways that information can be contributed to the knowledge base:

- [Contributing using the issue tracker.](#)
- [Contributing directly via the code repo.](#)

All content contributions should start in the issue tracker, since discussion is easier there about concepts than in code review.

Contributing using the issue tracker

This is the easiest way to propose updates to content in the SOLVE-IT knowledge base.

- Visit the SOLVE-IT [issue tracker](#).
- Select one of the templates. We offer 'STANDARD' or 'LITE' templates for content contributions:
 - STANDARD asks for all the details needed to populate the content in a structured for.
 - LITE requests the minimal information needed, with a free text 'notes' section for additional information.
- Submit the suggestion as a github issue for review.

Contributing directly via the code repo

- All contributions to the project must be submitted via pull request.
- Ensure that your pull request addresses a specific issue or feature request. If none exists, please open a new issue first to discuss the updates.
- Follow the [GitHub Flow](#) workflow:
 - i. Create a new branch from `main`.
 - ii. Make your changes and commit them with descriptive commit messages.

<https://github.com/SOLVE-IT-DF/solve-it/blob/main/CONTRIBUTING.md>

SOLVE-IT: contributing

SOLVE-IT-DF / solve-it

Q Type / to search

<> Code

Issues

36

Pull requests

4

Actions

Projects

Security

Insights

is:issue state:open

Q

Labels

Milestones

New issue

☐ Open

36

Closed

10

Author ▾

Labels ▾

Projects ▾

Milestones ▾

Assignees ▾

⇅ Newest ▾

☐

Possible duplication with M1004 and M1009

#99 · chrishargreaves opened 11 hours ago

1

☐

Remove legacy solveitcore.py

#94 · chrishargreaves opened 2 days ago

1

☐

Update CASE mapping for T1006 Digital sniffer dogs

solve-it content update

#84 · chrishargreaves opened 3 weeks ago

1

☐

Reword T1011 Evidence bags

solve-it content update

#83 · chrishargreaves opened 3 weeks ago

1

1

☐

Identifying and Locating Spoofed or Covert RF Devices

#80 · Jezza2026 opened last month

1

☐

Identifying and Locating Smartphones via RF Emissions

#79 · Jezza2026 opened last month

2

☐

Review labels needed for issue tracker

documentation

#74 · chrishargreaves opened on Jun 9



☐

Review weaknesses and mitigations related to T1007 and T1116 for consistency and de-duplication

solve-it content update

#73 · chrishargreaves opened on Jun 9

1

☐

Make case consistent in the excel output for INCOMP etc.

bug

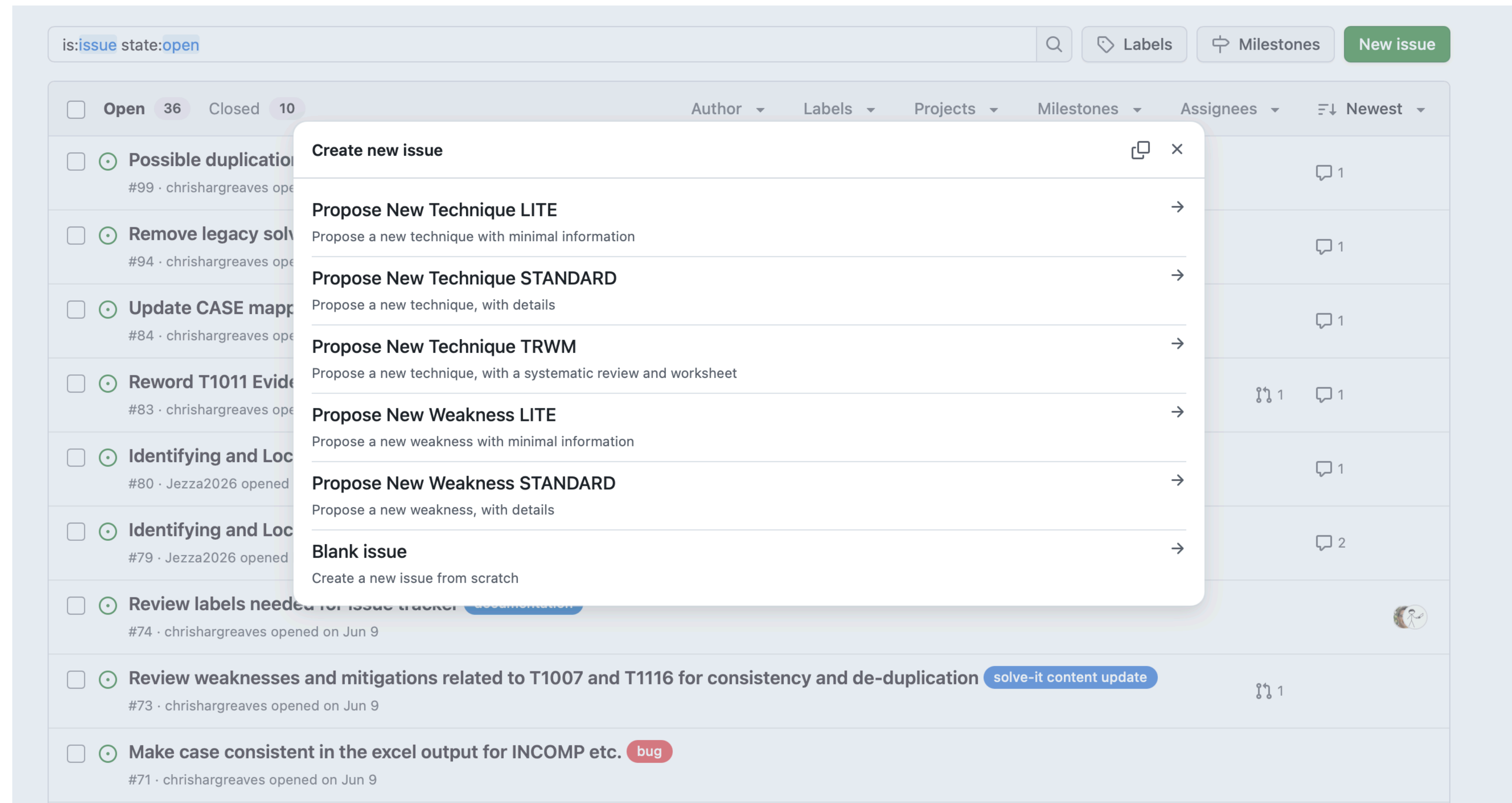
#71 · chrishargreaves opened on Jun 9

Microsoft

Display a menu

SOLVE-IT: issue templates

- LITE
- STANDARD
- TRWM (experimental)



SOLVE-IT: issue templates: LITE

- LITE - intended as a very lightweight submission.
- Good for capturing that a technique or weakness is missing without thorough details.
- Will take longer to get implemented as we may have to ask further questions or conduct research.

Create new issue in SOLVE-IT-DF/solve-it: Propose New Technique LITE

Add a title *

Create new technique: [technique name]

Add a description

WritePreview

HBI≡<>@↗↶↵

Technique name:
The name for the technique

Description:
A short description of the technique

Notes:
Any other notes that would be helpful in populating this technique later

Paste, drop, or click to add files

Assignee

Label

Project

Milestone

☐ Create more

Cancel

Create ↗ ↶

SOLVE-IT: issue templates: STANDARD

- STANDARD - has a more extensive template that includes all the detail we (or an implementer) would need to add this to the knowledge base.
- Includes all the fields discussed earlier
- Includes a link to the CASE ontology and CASE Issue Tracker to propose new classes if something is not represented.

**Create new issue**
Propose New Technique STANDARD · [Choose a different template](#)

Add a title *

Add a description

Write Preview H B I | [list icons] | [link icon] | [share icon] | [undo icon] | [redo icon]

Please populate these fields as best as you can:

Technique name:
The name for the technique (synonyms can be provided later)

Description:
A short description of the technique (extended details can be provided later).

Synonyms:
Other names that might be used for this technique.

Details:
A longer set of details can be provided to describe the technique.

Examples:
Examples can be provided e.g. cases where the technique can be used, or tools that offer the technique.

Weaknesses:
*Free text here

* Weakness 1
* Weakness 2
* ASTM Error
* Reference
* Mitigation 1
* Mitigation 2
* Mitigation 3

■ Mitigation references
...

CASE output entity:

- Refer to [CASE Ontology A-Z](#) to see if any suitable CASE entities exist to model the output of the technique.

Proposed CASE output entity (if none are suitable):

You can propose a CASE output entity here if you think it would be useful to model in CASE. You can also submit a proposed class directly via the [CASE Issue Tracker](#)

References

A list of references to support any details provided above. Harvard is used for now but we will move to bibtex representations in future.

SOLVE-IT: issue templates: TRWM

- TRWM (Technique Results Weakness Mitigation) - a systematic approach for populating a technique in the knowledge base.
- Trwm (/trʊm/), means 'heavy' in Welsh, and that is a fairly appropriate name. :)
- This is quite an involved process, but it is thorough.
- Supported by *SOLVE-IT Helper for Weakness And Mitigation Analysis* (SHWAMA) worksheets.
- Still experimental, but there are a few worked examples and has been useful so far.

Create new issue in SOLVE-IT-DF/solve-it: Propose New Technique TRWM

Add a title *

Title

Add a description

WritePreview

HBI|≡<>@↗↶

This an experimental systematic approach for enumerating weaknesses and mitigations for a technique.

TRWM is the approach that systematically reviews in turn the Technique, the technique Results, the Weaknesses, then the Mitigations. TRWM-A is the version of this method that uses ASTM E3016-18 error classifications to assist with weakness enumeration.

SHWAMA worksheets (SOLVE-IT Helper for Weakness And Mitigation Analysis) is a Google Sheet template that walks through the TRWM process with prompts, placeholder text, and linked worksheets that result in a summary sheet that can be used to implement the technique, weakness and mitigation content within the knowledge base.

An exercise on using the TRWM method and SHWAMA worksheets is available [in the SOLVE-IT education repository](https://github.com/SOLVE-IT-DF/solve-it-education).

The latest SHWAMA worksheet template is available [here] (https://docs.google.com/spreadsheets/d/1DRHCP7zAHBfz2TZ8yf3fhcc-bRZD25BT2oUAo1ZN6R4), which can be duplicated and used to work through the technique being studied.

Please provide a link to the completed SHWAMA worksheet below:

<add link here>

☐ Create more

Cancel

Create ↗ ↶

SOLVE-IT: issue templates: SHWAMA

SOLVE-IT:TRWM-A:SHWAMA V2.2 RC2

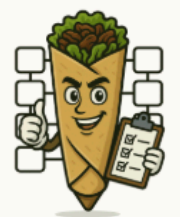
File Edit View Insert Format Data Tools Extensions Help

Search Menus

Undo Redo Print Copy Paste 100% \$ % .0 .00 123 Arial - 10 + B I A

Grid Borders Conditional Formatting Styles Font Color Background Color Link Unlink Chart Filter Sort Sum

A1:D1 fx

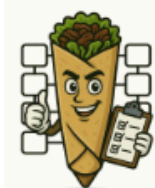
	A	B	C	D	E
1					
2		SOLVE-IT Helper for Weakness And Mitigation Analysis (SHWAMA) An implementation of the TRWM (Technique Results Weaknesses Mitigations) (ASTM-based) (TRWM-A) approach			
3	Information This is a workbook that helps perform a systematic review of a technique in digital forensics and to create or update content for inclusion in the SOLVE-IT knowledge base. It consists of working through four stages (TRWM) to populate content: 1. Technique 2. Results (from using the technique) 3. Weaknesses 4. Mitigations This specific implementation uses TRWM-A, which refers to the use of the ASTM E3016-18 error classifications to help iterate through potential weaknesses. Go to populating the Technique worksheet...				
4					
5					
6					
7	v2.2				

Add 1000 more rows at the bottom

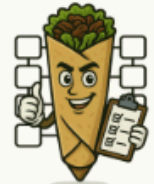
+ ≡ Information 1_Technique 2_Results 3a_Weakness Prompts (ASTM) 3b_Weakness (aggregate) (ASTM) 4_Mitigations 4b_Mitigation Summary Summary < >

Template: <https://docs.google.com/spreadsheets/d/1DRHCP7zAHBfz2TZ8yf3fhcc-bRZD25BT2oUAo1ZN6R4>

SOLVE-IT: issue templates: SHWAMA

	A	B	C	D	E
1					
2		SOLVE-IT Helper for Weakness And Mitigation Analysis (SHWAMA) <i>An implementation of the TRWM (Technique Results Weaknesses Mitigations) (ASTM-based) (TRWM-A) approach</i>			
3	Explanation				
4	This phase is to document the nature of the technique, and provide a description and examples of use. Ideally the name and description of the technique would come from an academic, government or industry source, but some techniques do not have a clean definition in literature. Additional explanatory notes for each field are provided on the right hand side.				
5		Field	Contents	Notes	
6		SOLVE-IT repo link:	https://github.com/SOLVE-IT-DF/solve-it/blob/main/data/techniques/T1073.json	You can copy the link from the project repo address bar e.g. https://github.com/SOLVE-IT-DF/solve-it/blob/main/data/techniques/T1000.json	
7		Technique ID:	T1073	This is only needed if this is an existing technique, ignore this if a new technique is being proposed.	
8		Technique name:	Calendar app examination	Current or proposed name for the technique.	
9		Technique description:	The analysis of built in or third-party calendar applications.	A description or definition of the technique. This will ideally be backed by literature from academia, government or industry.	
10		Synonyms:		Many techniques in digital forensics are referred to by multiple names. Synonyms for the technique can be added here to assist SOLVE-IT users in finding the relevant technique.	
11		Technique details:	This technique has the potential to associate the owner of the smartphone to specific locations at specific times (adapted from Alghafli et al. 2011)	This field can be used to provide more details beyond the definition, to keep that definition concise.	
12		Examples:		Here examples can be provided that make use of the technique.	
13		References:	Alghafli, K.A., Jones, A. and Martin, T.A., 2011. Guidelines for the digital forensic processing of smartphones. 9th Australian Digital Forensics Conference, Edith Cowan University, Perth Western Australia, 5th -7th December 2011 Metz (2009) E-mail and appointment falsification analysis, Forensic Focus, https://www.forensicfocus.com/articles/e-mail-and-appointment-falsification-analysis	Where possible, please provide references to support the details provided above. References should not be added just because they are about a topic, but should have meaningful implications in terms of explaining a technique. For large references, consider supplying the page or chapter number if appropriate. Harvard referencing is used for now, but we will move to bibtex representations in future.	
+ ≡ Information ▾ 1_Technique ▾ 2_Results ▾ 3a_Weakness Prompts (ASTM) ▾ 3b_Weakness (aggregate) (ASTM) ▾ 4_Mitigations ▾ 4b_Mitigation Summary ▾ Summary					

SOLVE-IT: issue templates: SHWAMA

	A	B	C	D	E	F
1						
2		SOLVE-IT Helper for Weakness And Mitigation Analysis (SHWAMA) An implementation of the TRWM (Technique Results Weaknesses Mitigations) (ASTM-based) (TRWM-A) approach				
3	Explanation					
4	This phase is to identify the results that can emerge from using the specified digital forensic technique. These are referred to as: Digital Forensic Technique Results/Outputs (DFTRs). They will be used in the next stage to help enumerate potential weaknesses of using a digital forensic technique. These can be identified in multiple ways. * If the techqniue involves the extraction of artefacts, e.g. <i>photos app examination</i>, then it is useful to experiment with one or more of these types of app and note the data points that could be relevant to a forensic investigation, e.g. photo content, photo taken time, album names, list of photos in album. * It is also useful to look at forensic tools that already process this type of app and record the type of artefacts that are already extracted, e.g. latitude and longitude recorded in a photo. Record each of the potential results below. You can provide a short name, a description, and also consult the CASE Ontology a-z list of classes (linked below) to see if this type of result is already included in the ontology. If it is not, you can note a potential new CASE class in the 'Potential CASE class' column.					
	5	DFTR ID	Name	Description	CASE class (refer to a-z list)	Potential CASE class
	6	R1	Calendar accounts	Accounts that contain individual calendars	observable:Account	
	7	R2	Calendars	A calendar that contains calendar entries (events)	observable:Calendar	
	8	R3	Event	An entry in a calendar referring to a specific event at a time	observable:CalendarEntry	
9	R4	Event name	The name of the event		CalendarEntryName	
10	R5	Event timing	The timings of the event, start, end, duration, repeat status, 'all day' status		CalendarEventDateTime	
11	R6	Event location	The location of the event (physical, online meeting)	observable:location , observable:URL	OnlineMeetingID, OnlineMeetingPassword	
12	R7	Event invitees	The identifiers of people invited to the event	identity:Identity		
13	R8	Calendar settings	Settings associated with the calendar app (e.g. if alternative calendars are in use)			
14	R9					
15	R10					
16	R11					
17	R12					
18						
19	Go to Weakness prompts worksheet...					
20						

Add

1000 more rows at the bottom

SOLVE-IT: issue templates: SHWAMA

3	Explanation									
4	This phase is to systematically review the potential results identified in the previous stage, and to systematically consider what weaknesses could occur. This worksheet will auto-populate with the Digital Forensics Technique Results (DFTRs) from the previous sheet and map them against potential output error classes. There are different versions of this mapping possible, for example against the desirable digital evidence properties of authentic, accuracy, complete, but for improved granularity, this version of the worksheet uses TRWM-A, which uses the ASTM E3016-18 error classifications. These are: * Incompleteness (INCOMP) * Inaccuracy - Existence (INAC-EX) * Inaccuracy - Alteration (INAC-ALT) * Inaccuracy - Association (INAC-AS) * Inaccuracy - Corruption (INAC-COR) * Misinterpretation (MISINT) You can consider each result in turn, and determine if any of the error classes can occur for that result, and if so what the nature of the error or weakness is. Describe the nature of the weakness in the spaces provided. Example prompts for each error class are included in each section to help think about what an error of that type looks like for a specific result. Please note that this version can handle a maximum of 30 weaknesses, but this will be extended in a future version.									
5										
6	When complete, you can go to the Mitigations worksheet									
7	INCOMPINAC-EXINAC-ALTINAC-ASINAC-CORMISINTError class summary									
8	R1	Calendar accounts	Accounts that contain individual calendars	Note, these are precompleted to align with the main error class, but additional error classes sometimes apply to a weakness and can be marked with an X.						
9	INCOMP	Prompts: failure to recover live artefacts, failure to recover deleted artefacts, other reasons why an artefact might be missed?								
10		Failure to recover live calendar accounts			X					INCOMP
11		Failure to recover deleted but recoverable calendar accounts			X					INCOMP
12					X					INCOMP
13					X					INCOMP
14					X					INCOMP
15	INAC-EX	Prompts: presenting an artefact for something that does not exist								
16		Presentation of an account that does not exist				X			X	INAC-EX, MISINT
17						X				INAC-EX
18						X				INAC-EX
19						X				INAC-EX
20						X				INAC-EX
21	INAC-ALT	Prompts: modifying the content of some digital data								
22							X			INAC-ALT
23							X			INAC-ALT
24							X			INAC-ALT
25							X			INAC-ALT
26							X			INAC-ALT

Information

1_Technique

2_Results

3a_Weakness Prompts (ASTM)

3b_Weakness (aggregate) (ASTM)

4_Mitigations

4b_Mitigation Summary

Summary

<

>

SOLVE-IT: issue templates: SHWAMA

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q		
228															X	MISINT			
229															X	MISINT			
230		R7	Event invitees	The identifiers of people invited to the event						INCOMP	INAC-EX	INAC-ALT	INAC-AS	INAC-COR	MISINT				
231		INCOMP	Prompts: failure to recover live artefacts, failure to recover deleted artefacts, other reasons why an artefact might be missed?																
232				Failure to recover event attendee information						X							INCOMP		
233				Failure to recover event attendee acceptance detail						X							INCOMP		
234										X							INCOMP		
235										X							INCOMP		
236										X							INCOMP		
237		INAC-EX	Prompts: presenting an artefact for something that does not exist, ...																
238				Presenting an invitee not associated with the event							X						INAC-EX		
239											X						INAC-EX		
240											X						INAC-EX		
241											X						INAC-EX		
242											X						INAC-EX		
243		INAC-ALT	Prompts: modifying the content of some digital data																
244												X					INAC-ALT		
245												X					INAC-ALT		
246												X					INAC-ALT		
247												X					INAC-ALT		
248												X					INAC-ALT		
249		INAC-AS	Prompts: presenting live data as deleted and vice versa																
250													X				INAC-AS		
251													X				INAC-AS		
252													X				INAC-AS		
253													X				INAC-AS		
254													X				INAC-AS		
255		INAC-COR	Prompts: could the process corrupt data, could the process fail to detect corrupt data?																
256														X			INAC-COR		
257														X			INAC-COR		
258														X			INAC-COR		
259														X			INAC-COR		
260														X			INAC-COR		
261		MISINT	Prompts: could results be presented in a way that encourages misinterpretation?																
262				Failure to display invitee event acceptance detail											X		MISINT		
263															X		MISINT		
264															X		MISINT		
265															X		MISINT		
266															X		MISINT		
267		R8	Calendar settings	Settings associated with the calendar app (e.g. if alternative calendars are in use)						INCOMP	INAC-EX	INAC-ALT	INAC-AS	INAC-COR	MISINT				
268		INCOMP	Prompts: failure to recover live artefacts, failure to recover deleted artefacts, other reasons why an artefact might be missed?																
269										X							INCOMP		
270										X							INCOMP		

SOLVE-IT: issue templates: SHWAMA



SOLVE-IT Helper for Weakness And Mitigation Analysis (SHWAMA)

An implementation of the TRWM (Technique Results Weaknesses Mitigations) (ASTM-based) (TRWM-A) approach

Explanation

There is nothing to edit on this sheet currently. It presents a summary of the weaknesses identified and their error classes.

Data on this sheet is used to help automatically populate other sheets.

Automatic Aggregated Weakness List	ASTM						Error class summary
	INCOMP	INAC-EX	INAC-ALT	INAC-AS	INAC-COR	MISINT	
Failure to recover live calendar accounts	X						INCOMP
Failure to recover deleted but recoverable calendar accounts	X						INCOMP
Presentation of an account that does not exist		X				X	INAC-EX, MISINT
Presenting a live account as deleted				X			INAC-AS
Presenting a deleted account as live				X			INAC-AS
Failing to display that an account was flagged as inactive						X	MISINT
Failure to recover live calendars	X						INCOMP
Failure to recover deleted but recoverable calendars	X						INCOMP
Presenting a calendar that does not exist		X					INAC-EX
Presenting a live calendar as deleted				X			INAC-AS
Presenting a deleted calendar as live				X			INAC-AS
Failure to recover deleted but recoverable events	X						INCOMP
Failure to recover events removed as archived or synced online	X						INCOMP
Failure to recover live events from a calendar	X						INCOMP
Presenting a calendar entry that does not exist		X					INAC-EX
Associating an event with the wrong calendar				X		X	INAC-AS, MISINT
Associating an event with the wrong account				X		X	INAC-AS, MISINT
Failure to detect modification to the event					X	X	INAC-COR, MISINT
Failure to recover event name	X						INCOMP
Presenting a calendar name that does not exist		X					INAC-EX
Missing events due to not taking into account event recurrences	X						INCOMP
Presenting an event at an incorrect time due to incorrect timezone parsing			X				INAC-ALT
Failure to recover event location information	X						INCOMP
Presenting a location not present in the event		X					INAC-EX
Failure to recover event attendee information	X						INCOMP
Failure to recover event attendee acceptance detail	X						INCOMP
Presenting an invitee not assoicated with the event		X					INAC-EX
Failure to display invitee event acceptance detail						X	MISINT
Failure to display alternative calendars in forensic tool, when they were in						X	MISINT
	-	-	-	-	-	-	

+

≡

Information ▾

1_Technique ▾

2_Results ▾

3a_Weakness Prompts (ASTM) ▾

3b_Weakness (aggregate) (ASTM) ▾

4_Mitigations ▾

4b_Mitigation Summary ▾

Summary |

<

>

SOLVE-IT: issue templates: SHWAMA



SOLVE-IT Helper for Weakness And Mitigation Analysis (SHWAMA)

An implementation of the TRWM (Technique Results Weaknesses Mitigations) (ASTM-based) (TRWM-A) approach

Explanation

This phase is to now think about the mitigations for the weaknesses that have been identified.

Aside from considering each weakness in turn, a systematic method of mitigation enumeration does not yet exist. However, broadly speaking some categories are:

- checking results manually or with another tool
- testing
- using an alternative approach
- using a complimentary approach
- checking for supporting or contradictory information

The weaknesses have been auto populated below and the relevant error classes summarised as a reminder.

WARNING: The weaknesses are auto populated from earlier data, so if you go back and add or remove weaknesses, mitigations may go out of sync.

Note that at present, this prototype can handle at most 30 weaknesses, but will be extended in future.

Total weaknesses: 29

[When complete, go to the Compact Summary worksheet](#)

Weaknesses (corresponding results are deliberately not repeated here to ensure that weakness description make sense as stand alone descriptions)	Reminder of error classes of weakness	Proposed mitigation descriptions
Failure to recover live calendar accounts	INCOMP	Tool testing for extraction of calendar account information
		Manual verification of relevant data
		Dual tool verification
		Tool testing for extraction of calendar account information
		Manual verification of relevant data
		Dual tool verification

+

≡

Information ▾

1_Technique ▾

2_Results ▾

3a_Weakness Prompts (ASTM) ▾

3b_Weakness (aggregate) (ASTM) ▾

4_Mitigations ▾

4b_Mitigation Summary ▾

Summary

<

>

SOLVE-IT: issue templates: SHWAMA

[illegible]

SOLVE-IT: issue templates: SHWAMA

A	B	C	D	E	F	G	H	I	J	K	L	
Generated using TRWM/SHWAMA	v2.2											
Technique name:	Calendar app examination											
Technique description:	The analysis of built in or third-party calendar applications.											
Synonyms:												
Details:	This technique has the potential to associate the owner of the smartphone to specific locations at specific times (adapted from Alghaffi et al. 2011)											
Examples:												
CASE output classes:	observable:Account, observable:Calendar, observable:CalendarEntry, observable:location, observable:URL, identity:Identity											
Proposed CASE classes:	CalendarEntryName, CalendarEventDateTime, OnlineMeetingID, OnlineMeetingPassword											
References:	Alghaffi, K.A., Jones, A. and Martin, T.A., 2011. Guidelines for the digital forensic processing of smartphones. 9th Australian Digital Forensics Conference, Edith Cowan University, Perth Western Australia, 5th -7th December 2011 Metz (2009) E-mail and appointment falsification analysis, Forensic Focus, https://www.forensicfocus.com/articles/e-mail-and-appointment-falsification-analysis											
Weaknesses and mitigations:		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	
Failure to recover live calendar accounts	INCOMP	Tool testing for extraction of calendar account information	Manual verification of relevant data	Dual tool verification								
Failure to recover deleted but recoverable calendar accounts	INCOMP	Tool testing for extraction of calendar account information	Manual verification of relevant data	Dual tool verification								
Presentation of an account that does not exist	INAC-EX, MISINT	Tool testing for extraction of calendar account information	Manual verification of relevant data	Dual tool verification								
Presenting a live account as deleted	INAC-AS	Tool testing for extraction of calendar account information	Manual verification of relevant data	Dual tool verification								
Presenting a deleted account as live	INAC-AS	Tool testing for extraction of calendar account information	Manual verification of relevant data	Dual tool verification								
Failing to display that an account was flagged as inactive	MISINT	Ensure 'active status' of calendar accounts is viewed	Manual check of calendar account status									
Failure to recover live calendars	INCOMP	Tool testing for extraction of calendars	Manual verification of relevant data	Dual tool verification								
Failure to recover deleted but recoverable	INCOMP	Tool testing for extraction of	Manual verification of relevant data	Dual tool verification								

+

≡

▼

1_Technique ▼

2_Results ▼

3a_Weakness Prompts (ASTM) ▼

3b_Weakness (aggregate) (ASTM) ▼

4_Mitigations ▼

4b_Mitigation Summary ▼

Summary (compact) ▼

<

>

SOLVE-IT-DF

🔍 Type / to search

📖 Overview

📁 Repositories 4

📁 Projects

📁 Packages

★ Stars



Follow

7 followers · 0 following

solve-it

The SOLVE-IT knowledge base for digital forensics

● Python ☆ 39 🍷 5

Public

solve-it-applications-ai-review

An application of SOLVE-IT to review AI applicability to DF techniques

● TeX ☆ 2 🍷 1

Public

solve-it-examples

Practical examples using the SOLVE-IT knowledge base.

Public

solve-it-education

Material that can be used for teaching using SOLVE-IT

Public

Learn how we count contributions

Less More

2025

2024

SOLVE-IT: education repository

SOLVE-IT Education Resources

Introduction

This repository is to host material that can be used in SOLVE-IT.

This is in early development.

- A one page primer introducing SOLVE-IT
- Class exercises

Presentations and other material

One Page Primer

This one page primer was created

[SOLVE-IT One Page Primer v1.0](#)

Class Exercises

These are exercises that have been designed to be used in a classroom. They may be adapted to your needs and enhancements can be suggested here.

Reviewing class exercise examination workflows

This exercise is designed as an add-on to the end of a practical examination of a data source e.g. a disk image or phone extraction. Students will need to consider the tools and processes used, dissect them into component parts and map them against SOLVE-IT techniques.

[SOLVE-IT Exercise_ Reviewing class exercise examination workflows.pdf](#)

Tool testing dataset creation

This exercise is designed to encourage students to think about how tools can go wrong, and how we can capture that in a dataset, and use that dataset to test tools.

[SOLVE-IT Exercise_ Tool testing dataset creation.pdf](#)

Weakness Enumeration with TRWM and SHWAMA

This exercise will ask students to take a technique from SOLVE-IT, or one that is missing, and to systematically enumerate the weaknesses and mitigations associated with the technique. TRWM is the general approach of systematically considering: Technique, Result, Weaknesses and Mitigations.

[SOLVE-IT Exercise_ Weakness Enumeration with TRWM and SHWAMA.pdf](#)

Questions?

chris@dfrrws.org